

Supplementary Material

Differentiable Interval Bottlenecks for Interpretable Anomaly Detection in Numerical Data

Anonymous Author(s)

Submission to IEEE ICDM 2026, Double-Blind Review

Abstract—This supplement is the complete experimental record behind the main paper, together with the material deferred there for space. It is organized as a roadmap: Section I details the datasets, baselines, and evaluation protocol; Section II the per-dataset robustness ablations; Section III the interpretability evaluation suite (label-free ranking, faithfulness perturbation, conciseness, and human-readability); Section IV the complexity and scalability profile; and Section V the *complete* per-dataset ROC–AUC and AUPRC for all 48 ADBench datasets and 23 methods (mean $\pm$ std over 10 seeds, with ranks and top-3 highlighting), plus the per-scale ROC–AUC critical-difference diagrams. The certified-Lipschitz analysis and the theorem-to-practice bridge figure live in the main paper (Sec. VI.E). All proofs are in the appendix of the main paper. Code and this document: <https://github.com/DiffInt/diffint>.

I. DATASETS AND EXPERIMENTAL PROTOCOL

A. Datasets

We evaluate on the 48 numerical datasets of the ADBench benchmark, grouped by scale into *small* (12), *medium* (15), *large* (11), and *high-dimensional* (10). Tables I–IV list, per dataset, the number of samples, features, and outliers, the outlier percentage, and the batch size used by DIFFINT (64 for $n \leq 10k$, 512 for $10k < n \leq 20k$, 1024 otherwise). The suite spans four orders of magnitude in sample size ($n \in [80, 6.2 \times 10^5]$) and from 3 to 1,555 features, so it stresses both data-scarce and high-dimensional regimes.

TABLE I: Small-scale datasets.

Dataset	Samples	Feat.	#Out.	Out. %	Batch
4_breastw	683	9	239	34.99	64
14_glass	214	7	9	4.21	64
15_Hepatitis	80	19	13	16.25	64
18_Ionosphere	351	33	126	35.90	64
21_Lymphography	148	18	6	4.05	64
29_Pima	768	8	268	34.90	64
37_Stamps	340	9	31	9.12	64
39_vertebral	240	6	30	12.50	64
42_WBC	223	9	10	4.48	64
43_WDBC	367	30	10	2.72	64
45_wine	129	13	10	7.75	64
46_WPBC	198	33	47	23.74	64

B. Protocol, baselines, and metrics

Protocol. DIFFINT is trained *semi-supervised* on inliers only (label 0), with a 40% test split and 10 random seeds, under a single $[-1, 1]$ -normalized configuration shared across all datasets ($\text{lr } 5 \times 10^{-5}$, $K=200$, $\tau=0.1$, $\rho=0.999$, 1000 epochs, Adam); no hyper-parameter is tuned per dataset. **Baselines.** We

TABLE II: Medium-scale datasets.

Dataset	Samples	Feat.	#Out.	Out. %	Batch
2_annthyroid	7,200	6	534	7.42	64
6_cardio	1,831	21	176	9.61	64
7_Cardiotocography	2,114	21	466	22.04	64
12_fault	1,941	27	673	34.67	64
19_landsat	6,435	36	1,333	20.71	64
20_letter	1,600	32	100	6.25	64
27_PageBlocks	5,393	10	510	9.46	64
28_pendigits	6,870	16	156	2.27	64
30_satellite	6,435	36	2,036	31.64	64
31_satimage-2	5,803	36	71	1.22	64
38_thyroid	3,772	6	93	2.47	64
40_vowels	1,456	12	50	3.43	64
41_Waveform	3,443	21	100	2.90	64
44_Wilt	4,819	5	257	5.33	64
47_yeast	1,500	8	20	1.33	64

TABLE III: Large-scale datasets.

Dataset	Samples	Feat.	#Out.	Out. %	Batch
1_ALOI	49,534	27	1,508	3.04	1,024
8_celeba	202,599	39	4,547	2.24	1,024
10_cover	286,048	10	2,747	0.96	1,024
11_donors	619,326	10	36,710	5.93	1,024
13_fraud	284,807	29	492	0.17	1,024
16_http	567,498	3	2,211	0.39	1,024
22_magic.gamma	19,020	10	6,688	35.16	512
23_mammography	11,183	6	260	2.32	512
32_shuttle	49,097	9	3,511	7.15	1,024
33_skin	245,057	3	50,859	20.75	1,024
34_smtp	95,156	3	30	0.03	1,024

TABLE IV: High-dimensional datasets.

Dataset	Samples	Feat.	#Out.	Out. %	Batch
3_backdoor	95,329	196	2,329	2.44	1,024
5_campaign	41,188	62	4,640	11.27	1,024
9_census	299,285	500	18,568	6.20	1,024
17_InternetAds	1,966	1,555	368	18.72	64
24_mnist	7,603	100	700	9.21	64
25_musk	3,062	166	97	3.17	64
26_optdigits	5,216	64	150	2.88	64
35_SpamBase	4,207	57	1,679	39.91	64
36_speech	3,686	400	61	1.65	64
48_arrhythmia	452	274	66	14.60	64

compare against 22 baselines from the ADBench/TCCM code-base: classical detectors (ABOD, LOF, PCA, KPCA, OCSVM, ECOD, CBLOF, KDE, GMM, MCD, HBOS, IForest, LMDD, QMCD, Sampling) and deep/inductive ones (AutoEncoder, VAE, DIF, LUNAR, SO/MO_GAAL, and the flow-matching model TCCM); together with DIFFINT this gives the 23

TABLE V: Input normalization. Mapping into a bounded symmetric domain is what matters; among bounded/standardized choices $[-1, 1]$ is the most accurate, within 1.5 points of the others.

Dataset	none	$[0, 1]$	standard	$[-1, 1]$
45_wine	0.990	0.932	0.883	0.950
21_Lymphography	0.990	0.998	0.996	0.994
46_WPBC	0.510	0.543	0.541	0.584
14_glass	0.896	0.886	0.877	0.876
42_WBC	0.990	0.989	0.990	0.998
39_vertebral	0.335	0.498	0.511	0.477
37_Stamps	0.924	0.921	0.929	0.953
Mean	0.805	0.824	0.818	0.833

TABLE VI: Training contamination: ROC–AUC vs. fraction of anomalies injected into the “inlier” training set. Degradation is graceful.

Dataset	0%	5%	10%	20%
45_wine	0.950	0.659	0.645	0.624
21_Lymphography	0.994	0.978	0.970	0.973
46_WPBC	0.584	0.565	0.570	0.520
14_glass	0.876	0.755	0.728	0.726
42_WBC	0.998	0.979	0.982	0.976
39_vertebral	0.477	0.458	0.432	0.456
37_Stamps	0.953	0.865	0.816	0.766
Mean	0.833	0.751	0.735	0.720

methods ranked in the tables. All methods share the same $[-1, 1]$ normalization. **Metrics.** We report ROC–AUC (area under the ROC curve) and AUPRC (average precision); both are threshold-free and higher-is-better. **Missing runs.** A few baselines did not complete on the largest datasets within budget; each such run is floored to the worst rank (the number of methods) on that dataset, the standard conservative convention used in the main paper. **How to read the result tables.** Every cell is $\text{mean}_{\pm\text{std}}(\text{rank})$ over the seeds; per column we highlight **rank 1**, **rank 2**, and **rank 3**.

II. ROBUSTNESS ABLATIONS

We expand the main-paper ablations to per-dataset numbers on seven small datasets ($n \geq 100$; $K=200$, 3 seeds; ROC–AUC), probing two design choices; Hepatitis ($n=80$) is excluded as too small for a stable inlier-only test split. **Normalization** (Table V): mapping into a bounded symmetric domain is what matters. Dropping it (“none”) is the only choice that collapses on several datasets, while among bounded/standardized choices $[-1, 1]$ is the most accurate, within 1.5 points of $[0, 1]$ and StandardScaler; we adopt it because it makes the theory transfer (main paper, Sec. III-B) and it is also the top performer. **Training contamination** (Table VI): injecting anomalies into the “inlier” training set degrades accuracy *gracefully* rather than catastrophically, since the EMA support averages over batches so a minority of contaminating points cannot dominate the learned intervals.

TABLE VII: Faithfulness perturbation test: score rise from crossing the top-LFI feature boundaries (Δ_{top}) vs. random features (Δ_{rand}), their ratio, and the fraction of inliers with $\Delta_{\text{top}} > \Delta_{\text{rand}}$.

Dataset	Δ_{top}	Δ_{rand}	ratio	% top>rand
14_glass	0.200	0.134	1.48	71%
42_WBC	0.094	0.074	1.26	62%
18_Ionosphere	0.105	0.078	1.34	64%
29_Pima	0.171	0.100	1.70	71%
6_cardio	0.119	0.066	1.80	79%
Mean	0.138	0.091	1.52	69%

TABLE VIII: Conciseness and human-readability proxies ($K=200$, 3 seeds). #feat_{80%}: features (ranked by per-feature error) explaining 80% of a flagged point’s score; active/unit: mean number of active features ($\bar{I}_{kj} < 0.9$) per top interval unit; width: mean active-interval width as a fraction of the $[-1, 1]$ range.

Dataset	d	#feat _{80%}	active/unit	width
14_glass	9	4.8	6.7	0.70
42_WBC	30	16.0	24.8	0.69
18_Ionosphere	33	17.9	29.8	0.70
29_Pima	8	4.6	5.3	0.70
6_cardio	21	11.0	17.2	0.70
Mean	20	10.8	16.8	0.70

III. INTERPRETABILITY AS EVIDENCE

We support the two interpretability claims of the main paper with quantitative measurements ($K=200$, 3 seeds; Table VII).

(1) **Label-free ranking (LFI).** We rank (unit, feature) constraints by the label-free importance $\text{LFI}_{kj} = s_{kj} \cdot (1 - \Delta_{kj}/\text{range}_j) \cdot \text{stab}_{kj}$ (support $\times$ tightness $\times$ stability), using only quantities the trained model already maintains. This is the only ranking we use in the main paper (Sec. V); it requires no anomaly label and is computable in closed form from the model parameters and the EMA support buffer.

(2) **Faithfulness perturbation.** For each test inlier we perturb its top-LFI features to *cross* their interval boundaries and measure the anomaly-score rise Δ_{top} , against perturbing the same number of random features Δ_{rand} . The top-ranked constraints raise the score $1.5\times$ more on average (on 69% of inliers; Table VII), so the LFI-highlighted constraints *causally* drive the score rather than merely correlating with it.

(3) **Conciseness and readability.** For a flagged point, the per-feature reconstruction error concentrates: on average $\approx$ half the features (10.8 of 20; Table VIII) account for 80% of the anomaly score, so a short shortlist suffices to explain it. The interval units themselves are *soft and wide* (mean width 0.70 of the feature range, and a unit softly constrains most coordinates by the membership criterion of Def. 1), which is precisely why we present intervals as *candidate constraints* rather than crisp rules and surface only the top- k by LFI ($k=2$ in the main-paper read-out) rather than a whole unit.

IV. COMPLEXITY AND SCALABILITY

Practical deployability matters as much as accuracy, so we summarize the resource profile of DIFFINT. Let n be the number of training points, d the feature count, K the number of interval units, B the batch size, E the epochs, and H the (fixed) decoder hidden width; the decoder is an MLP whose size is independent of n .

Parameter count. The interval bottleneck stores a centre and a softplus half-width per (unit, feature): $2Kd$ parameters, linear in K and d and *independent of n* . The decoder adds a compact MLP ($O(KH + Hd)$ weights). The total stays MLP-class (10^5 – 10^6 weights even at $d=1,555$, InternetAds), far below typical deep tabular detectors. Table IX reports the $2Kd$ bottleneck size and the matched-budget training time across scales.

Training complexity. A minibatch computes Bd sigmoid pairs for the soft memberships, an $O(BKd)$ aggregation (a per-unit log-sum over features plus an $O(K)$ softmax), and one decoder forward/backward; an epoch is thus $O(nKd)$ plus the n -independent decoder cost, with *no* pairwise, kernel, or neighbour computation. Total training is $O(EnKd)$.

Runtime vs. baselines. DIFFINT uses one forward pass per sample, unlike memory-based detectors (e.g. LUNAR) whose k NN queries grow with n . Empirically (main paper, Table II, matched 1000-epoch budget) DIFFINT trains one to two orders of magnitude faster than the binary differentiable miner BinaPs, faster than LUNAR on most datasets, and scales to the largest ADBench sets ($n \approx 6.2 \times 10^5$).

Inference and memory footprint. Scoring a point is a single forward pass: $O(Kd)$ memberships, the aggregation, the decoder, then the MAE. DIFFINT is *inductive*: it keeps only the trained parameters, not the training set, so inference latency and memory are constant in n , whereas transductive/memory-based detectors must retain or query the data. The training footprint is the parameters ($O(Kd)$), one batch of memberships ($O(BKd)$ activations), and the $K \times d$ EMA support buffer ($O(Kd)$); none depends on n . At inference only the $O(Kd)$ parameters and a single $K \times d$ membership tensor are held.

The K knob. K trades capacity for cost linearly. ROC–AUC saturates after a few units (main paper, Fig. 4), so K can be reduced well below the default under tight budgets with little accuracy loss; we keep $K=200$ as one untuned default across all 48 datasets.

TABLE IX: Resource profile across scales ($K=200$). Bottleneck parameters ($2Kd$) are linear in d and independent of n ; training time (DIFFINT, matched 1000-epoch budget) is reproduced from Table II of the main paper and grows with n and d .

Dataset	n	d	bottleneck $2Kd$	train (s)
6_cardio	1,831	21	8,400	3.0
31_satimage-2	5,803	36	14,400	54.6
26_optdigits	5,216	64	25,600	60.1
48_arrhythmia	452	274	109,600	30.2
32_shuttle	49,097	9	3,600	193.5

V. COMPLETE DETECTION ACCURACY

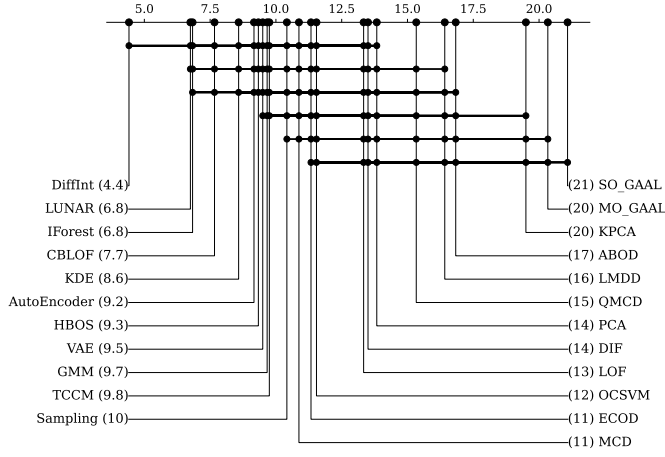
This section reports the full per-dataset detection accuracy underlying the critical-difference analysis of the main paper: ROC–AUC (Sec. V-A) and average precision (Sec. V-B). Across both metrics DIFFINT sits in the leading group on every scale and is most often first on small/medium/large data; in high dimensions it remains competitive with the strongest baselines.

A. ROC–AUC

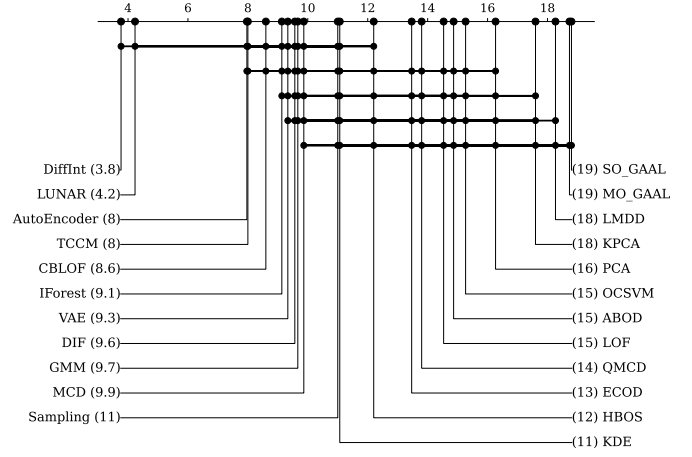
Tables X–XVIII give the per-dataset ROC–AUC. Figs. 1–2 show the per-scale critical-difference diagrams for ROC–AUC and AUPR (the overall diagrams are in the main paper). They support the regime-level claims of main paper Table I: DIFFINT attains the lowest mean rank in every (scale, metric) cell except high-dimensional AUPR, where LUNAR is marginally ahead but the two are statistically indistinguishable within the per-scale CD.

B. Average precision (AUPRC)

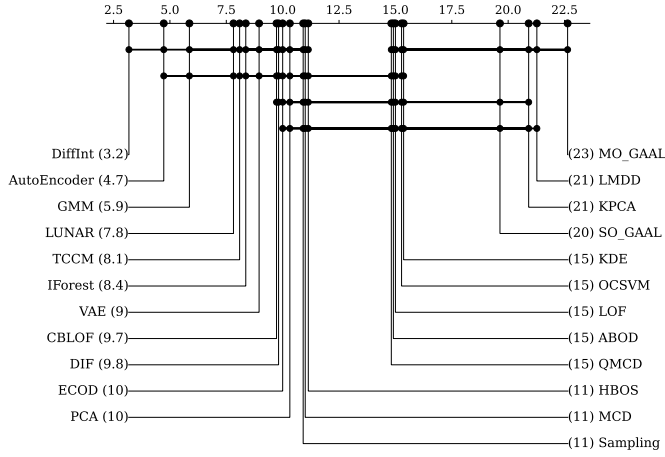
Tables XIX–XXVII give the per-dataset AUPRC under the same protocol and highlighting. AUPRC weights the rare anomaly class more heavily than ROC–AUC; the DIFFINT ranking is consistent across the two metrics, indicating the gains are not an artifact of the chosen measure.



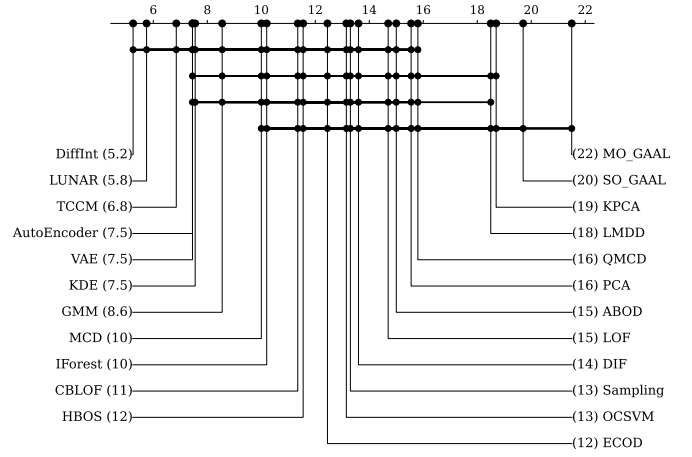
(a) Small ($n=12$).



(b) Medium ($n=15$).

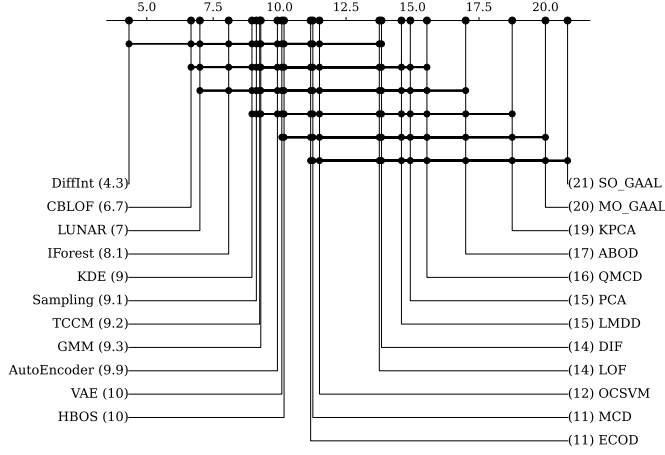


(c) Large ($n=11$).

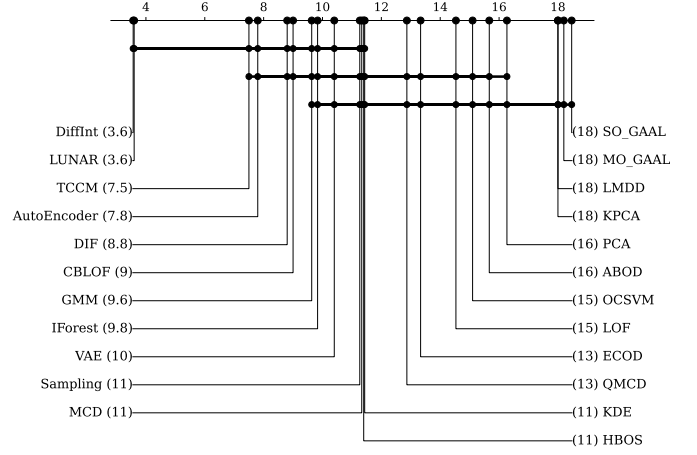


(d) High-dimensional ($n=10$).

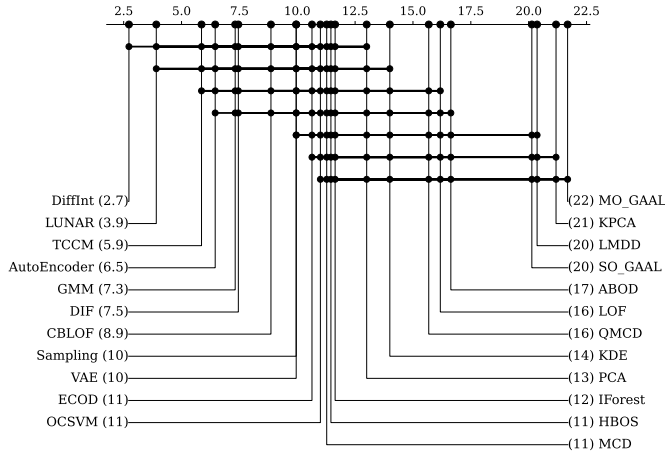
Fig. 1: Per-scale **ROC-AUC** critical-difference diagrams (missing runs floored to worst rank, bars indicate non-significant differences at $\alpha=0.05$). DIFFINT leads strictly on every scale; matches the “DIFFINT AUC” column of main paper Table I.



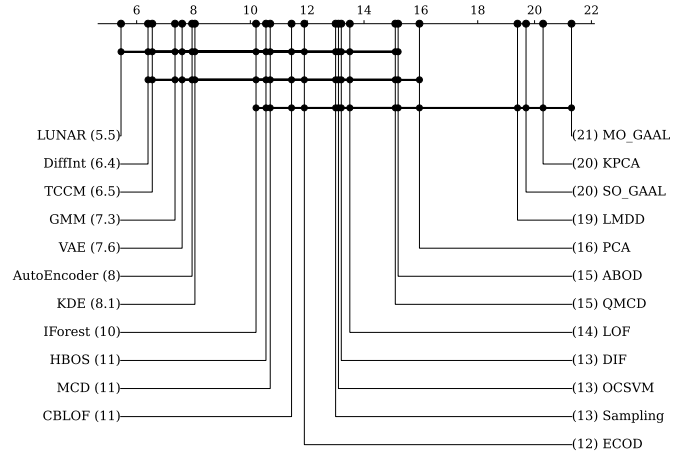
(a) Small ($n=12$).



(b) Medium ($n=15$).



(c) Large ($n=11$).



(d) High-dimensional ($n=10$).

Fig. 2: Per-scale **AUPR** critical-difference diagrams (same protocol as Fig. 1). DIFFINT leads strictly on small, medium, and large data; on high-dimensional data LUNAR’s mean rank (5.45) is slightly ahead of DIFFINT (6.40), but the gap is well inside the per-scale Nemenyi CD (≈ 11), i.e. a statistical tie. Matches the “DIFFINT AUPR” column of main paper Table I.

TABLE X: ROC–AUC, small datasets (1/2).

Model	4_breastw	14_glass	15_Hepatitis	18_Ionosphere	21_Lymphography	29_Pima
ABOD	0.3737 $\pm$ 0.0338(20)	0.8264 $\pm$ 0.0697(8)	0.6542 $\pm$ 0.1520(19)	0.9255 $\pm$ 0.0185(11)	0.9682 $\pm$ 0.0240(18)	0.6548 $\pm$ 0.0408(14)
AutoEncoder	0.9855 $\pm$ 0.0075(12)	0.7107 $\pm$ 0.0779(16)	0.8501 $\pm$ 0.0387(2)	0.9604 $\pm$ 0.0090(5)	0.9904 $\pm$ 0.0087(12)	0.7070 $\pm$ 0.0239(10)
CBLOF	0.9939 $\pm$ 0.0026(5)	0.8784 $\pm$ 0.0345(3)	0.8089 $\pm$ 0.0874(10)	0.9557 $\pm$ 0.0223(7)	0.9816 $\pm$ 0.0438(15)	0.7006 $\pm$ 0.0232(11)
DIF	0.8377 $\pm$ 0.0375(16)	0.8562 $\pm$ 0.0371(5)	0.7723 $\pm$ 0.0901(16)	0.9587 $\pm$ 0.0074(6)	0.8833 $\pm$ 0.0992(19)	0.6144 $\pm$ 0.0258(19)
DiffInt	0.9954 $\pm$ 0.0018(1)	0.8788 $\pm$ 0.0245(2)	0.8333 $\pm$ 0.0401(7)	0.9757 $\pm$ 0.0030(2)	0.9997 $\pm$ 0.0014(2)	0.7304 $\pm$ 0.0113(3)
ECOD	0.9932 $\pm$ 0.0029(7)	0.6911 $\pm$ 0.0797(17)	0.8402 $\pm$ 0.0550(3)	0.7745 $\pm$ 0.0221(17)	0.9990 $\pm$ 0.0022(4)	0.6374 $\pm$ 0.0239(16)
GMM	0.9863 $\pm$ 0.0055(11)	0.7508 $\pm$ 0.0624(14)	0.8018 $\pm$ 0.0785(13)	0.9664 $\pm$ 0.0071(3)	0.9873 $\pm$ 0.0092(13)	0.7202 $\pm$ 0.0217(7)
HBOS	0.9911 $\pm$ 0.0036(8)	0.7848 $\pm$ 0.0621(12)	0.8342 $\pm$ 0.0407(6)	0.7334 $\pm$ 0.0519(18)	0.9975 $\pm$ 0.0045(8)	0.7339 $\pm$ 0.0201(2)
IForest	0.9942 $\pm$ 0.0028(4)	0.7855 $\pm$ 0.0429(11)	0.8348 $\pm$ 0.0613(5)	0.9116 $\pm$ 0.0189(13)	0.9991 $\pm$ 0.0019(3)	0.7272 $\pm$ 0.0275(5)
KDE	0.9946 $\pm$ 0.0025(2)	0.7622 $\pm$ 0.0454(13)	0.7942 $\pm$ 0.0540(14)	0.9520 $\pm$ 0.0105(8)	0.9988 $\pm$ 0.0025(5)	0.7208 $\pm$ 0.0226(6)
KPCA	0.1983 $\pm$ 0.0825(21)	0.3975 $\pm$ 0.1372(20)	0.4379 $\pm$ 0.1895(21)	0.4606 $\pm$ 0.0632(21)	0.4137 $\pm$ 0.2537(20)	0.4976 $\pm$ 0.0274(21)
LMDD	0.7050 $\pm$ 0.1609(17)	0.3743 $\pm$ 0.1793(22)	0.5742 $\pm$ 0.1757(20)	0.6665 $\pm$ 0.1000(19)	0.9759 $\pm$ 0.0186(17)	0.6293 $\pm$ 0.0860(18)
LOF	0.4482 $\pm$ 0.0215(19)	0.8048 $\pm$ 0.0784(9)	0.8211 $\pm$ 0.0761(8)	0.8665 $\pm$ 0.0479(14)	0.9988 $\pm$ 0.0025(5)	0.6348 $\pm$ 0.0227(17)
LUNAR	0.9902 $\pm$ 0.0046(10)	0.8816 $\pm$ 0.0359(1)	0.7782 $\pm$ 0.0579(15)	0.9781 $\pm$ 0.0085(1)	0.9906 $\pm$ 0.0112(11)	0.7296 $\pm$ 0.0286(4)
MCD	0.9850 $\pm$ 0.0084(13)	0.7869 $\pm$ 0.0347(10)	0.8088 $\pm$ 0.0621(11)	0.9616 $\pm$ 0.0092(4)	0.9861 $\pm$ 0.0140(14)	0.7158 $\pm$ 0.0222(8)
MO_GAAL	0.0117 $\pm$ 0.0142(22)	0.3744 $\pm$ 0.2277(21)	0.3695 $\pm$ 0.1895(22)	0.3329 $\pm$ 0.0612(22)	0.0085 $\pm$ 0.0148(22)	0.3319 $\pm$ 0.0499(23)
OCSVM	0.9945 $\pm$ 0.0025(3)	0.4470 $\pm$ 0.1473(19)	0.8088 $\pm$ 0.0630(11)	0.8405 $\pm$ 0.0192(15)	0.9988 $\pm$ 0.0025(5)	0.6797 $\pm$ 0.0246(12)
PCA	0.9495 $\pm$ 0.0188(15)	0.6551 $\pm$ 0.0838(18)	0.8120 $\pm$ 0.0635(9)	0.7864 $\pm$ 0.0215(16)	0.9962 $\pm$ 0.0058(9)	0.6377 $\pm$ 0.0241(15)
QMCD	0.4800 $\pm$ 0.1746(18)	0.8423 $\pm$ 0.0490(6)	0.7150 $\pm$ 0.1246(18)	0.5509 $\pm$ 0.0455(20)	0.0884 $\pm$ 0.1204(21)	0.7417 $\pm$ 0.0239(1)
SO_GAAL	0.0110 $\pm$ 0.0116(23)	0.3392 $\pm$ 0.1817(23)	0.3134 $\pm$ 0.1751(23)	0.2823 $\pm$ 0.0611(23)	0.0049 $\pm$ 0.0108(23)	0.3350 $\pm$ 0.0491(22)
Sampling	0.9934 $\pm$ 0.0033(6)	0.8696 $\pm$ 0.0480(4)	0.7645 $\pm$ 0.0976(17)	0.9395 $\pm$ 0.0256(10)	0.9780 $\pm$ 0.0359(16)	0.7115 $\pm$ 0.0369(9)
TCCM	0.9677 $\pm$ 0.0195(14)	0.8334 $\pm$ 0.0914(7)	0.8394 $\pm$ 0.0708(4)	0.9400 $\pm$ 0.0158(9)	1.0000 $\pm$ 0.0000(1)	0.5562 $\pm$ 0.1001(20)
VAE	0.9903 $\pm$ 0.0045(9)	0.7148 $\pm$ 0.0655(15)	0.8600 $\pm$ 0.0425(1)	0.9206 $\pm$ 0.0155(12)	0.9932 $\pm$ 0.0068(10)	0.6714 $\pm$ 0.0323(13)

TABLE XI: ROC–AUC, small datasets (2/2).

Model	37_Stamps	39_vertebral	42_WBC	43_WDBC	45_wine	46_WPBC
ABOD	0.6954 $\pm$ 0.0525(19)	0.4214 $\pm$ 0.0847(18)	0.9297 $\pm$ 0.0301(18)	0.9143 $\pm$ 0.0504(18)	0.5828 $\pm$ 0.2278(20)	0.4897 $\pm$ 0.0563(19)
AutoEncoder	0.9217 $\pm$ 0.0398(10)	0.4644 $\pm$ 0.1041(12)	0.9841 $\pm$ 0.0220(12)	0.9937 $\pm$ 0.0055(2)	0.9364 $\pm$ 0.0499(6)	0.5178 $\pm$ 0.0481(11)
CBLOF	0.9457 $\pm$ 0.0235(4)	0.4711 $\pm$ 0.0742(10)	0.9961 $\pm$ 0.0039(2)	0.9860 $\pm$ 0.0099(14)	0.9517 $\pm$ 0.0376(2)	0.5280 $\pm$ 0.0407(9)
DIF	0.9426 $\pm$ 0.0215(5)	0.5781 $\pm$ 0.0895(3)	0.7821 $\pm$ 0.0531(19)	0.7227 $\pm$ 0.0519(19)	0.7978 $\pm$ 0.1078(17)	0.4971 $\pm$ 0.0485(18)
DiffInt	0.9516 $\pm$ 0.0116(1)	0.5274 $\pm$ 0.0429(4)	0.9973 $\pm$ 0.0050(1)	0.9741 $\pm$ 0.0076(16)	0.8852 $\pm$ 0.0594(13)	0.5652 $\pm$ 0.0222(1)
ECOD	0.9157 $\pm$ 0.0244(13)	0.4645 $\pm$ 0.0796(11)	0.9953 $\pm$ 0.0046(6)	0.9870 $\pm$ 0.0068(11)	0.8484 $\pm$ 0.0960(14)	0.5029 $\pm$ 0.0422(17)
GMM	0.9270 $\pm$ 0.0226(9)	0.4909 $\pm$ 0.0550(8)	0.9767 $\pm$ 0.0198(16)	0.9882 $\pm$ 0.0093(9)	0.9677 $\pm$ 0.0268(1)	0.5141 $\pm$ 0.0507(12)
HBOS	0.9375 $\pm$ 0.0146(7)	0.3982 $\pm$ 0.0437(21)	0.9896 $\pm$ 0.0093(9)	0.9890 $\pm$ 0.0063(7)	0.9296 $\pm$ 0.0419(9)	0.5483 $\pm$ 0.0532(5)
IForest	0.9485 $\pm$ 0.0202(2)	0.4426 $\pm$ 0.1017(15)	0.9948 $\pm$ 0.0049(7)	0.9903 $\pm$ 0.0056(4)	0.9393 $\pm$ 0.0441(5)	0.5287 $\pm$ 0.0451(8)
KDE	0.9309 $\pm$ 0.0174(8)	0.4096 $\pm$ 0.0856(20)	0.9960 $\pm$ 0.0045(3)	0.9864 $\pm$ 0.0075(13)	0.9336 $\pm$ 0.0339(7)	0.5603 $\pm$ 0.0511(2)
KPCA	0.4111 $\pm$ 0.1136(21)	0.5204 $\pm$ 0.0619(6)	0.3710 $\pm$ 0.2036(20)	0.2678 $\pm$ 0.0890(21)	0.3955 $\pm$ 0.2631(21)	0.4644 $\pm$ 0.0961(21)
LMDD	0.4454 $\pm$ 0.0266(20)	0.4355 $\pm$ 0.0811(16)	0.9769 $\pm$ 0.0615(15)	0.9941 $\pm$ 0.0033(1)	0.6987 $\pm$ 0.1469(18)	0.5125 $\pm$ 0.0619(14)
LOF	0.7229 $\pm$ 0.1204(18)	0.3515 $\pm$ 0.0717(23)	0.9759 $\pm$ 0.0198(17)	0.9871 $\pm$ 0.0088(10)	0.9115 $\pm$ 0.0592(12)	0.5326 $\pm$ 0.0635(7)
LUNAR	0.9473 $\pm$ 0.0248(3)	0.4484 $\pm$ 0.0817(13)	0.9794 $\pm$ 0.0237(14)	0.9935 $\pm$ 0.0055(3)	0.9478 $\pm$ 0.0439(3)	0.5536 $\pm$ 0.0417(3)
MCD	0.8368 $\pm$ 0.0255(17)	0.4872 $\pm$ 0.0529(9)	0.9807 $\pm$ 0.0230(13)	0.9682 $\pm$ 0.0141(17)	0.9444 $\pm$ 0.0521(4)	0.5277 $\pm$ 0.0682(10)
MO_GAAL	0.2743 $\pm$ 0.1774(22)	0.6858 $\pm$ 0.0735(1)	0.0050 $\pm$ 0.0063(23)	0.0074 $\pm$ 0.0074(22)	0.0320 $\pm$ 0.0388(22)	0.4373 $\pm$ 0.0912(22)
OCSVM	0.9027 $\pm$ 0.0260(14)	0.4294 $\pm$ 0.0843(17)	0.9960 $\pm$ 0.0045(3)	0.9887 $\pm$ 0.0057(8)	0.8410 $\pm$ 0.0944(16)	0.5137 $\pm$ 0.0344(13)
PCA	0.9197 $\pm$ 0.0195(12)	0.3973 $\pm$ 0.0598(22)	0.9907 $\pm$ 0.0091(8)	0.9868 $\pm$ 0.0058(12)	0.8450 $\pm$ 0.0888(15)	0.5096 $\pm$ 0.0543(15)
QMCD	0.9025 $\pm$ 0.0571(15)	0.4114 $\pm$ 0.0871(19)	0.3578 $\pm$ 0.1567(21)	0.5860 $\pm$ 0.1903(20)	0.6633 $\pm$ 0.1655(19)	0.5343 $\pm$ 0.0571(6)
SO_GAAL	0.2334 $\pm$ 0.1414(23)	0.6801 $\pm$ 0.0693(2)	0.0056 $\pm$ 0.0065(22)	0.0060 $\pm$ 0.0051(23)	0.0248 $\pm$ 0.0276(23)	0.4272 $\pm$ 0.0797(23)
Sampling	0.9210 $\pm$ 0.0424(11)	0.4449 $\pm$ 0.1259(14)	0.9871 $\pm$ 0.0271(11)	0.9817 $\pm$ 0.0095(15)	0.9328 $\pm$ 0.0487(8)	0.5505 $\pm$ 0.0742(4)
TCCM	0.8880 $\pm$ 0.0393(16)	0.5183 $\pm$ 0.0730(7)	0.9960 $\pm$ 0.0063(3)	0.9899 $\pm$ 0.0077(5)	0.9261 $\pm$ 0.0498(10)	0.4841 $\pm$ 0.0688(20)
VAE	0.9404 $\pm$ 0.0175(6)	0.5244 $\pm$ 0.0725(5)	0.9873 $\pm$ 0.0166(10)	0.9897 $\pm$ 0.0076(6)	0.9259 $\pm$ 0.0335(11)	0.5083 $\pm$ 0.0451(16)

TABLE XII: ROC–AUC, medium datasets (1/3).

Model	2_anthyroid	6_cardio	7_Cardiotocography	12_fault	19_landsat	20_letter
ABOD	0.7439 $\pm$ 0.0124(13)	0.5877 $\pm$ 0.0275(19)	0.4917 $\pm$ 0.0184(21)	0.6928 $\pm$ 0.0110(10)	0.5164 $\pm$ 0.0127(17)	0.8652 $\pm$ 0.0149(3)
AutoEncoder	0.8612 $\pm$ 0.0192(6)	0.9307 $\pm$ 0.0271(12)	0.7202 $\pm$ 0.0335(12)	0.7386 $\pm$ 0.0188(4)	0.5773 $\pm$ 0.0111(13)	0.8135 $\pm$ 0.0340(7)
CBLOF	0.6578 $\pm$ 0.0179(17)	0.9546 $\pm$ 0.0043(7)	0.7815 $\pm$ 0.0315(8)	0.7094 $\pm$ 0.0358(8)	0.6907 $\pm$ 0.0179(3)	0.7755 $\pm$ 0.0370(9)
DIF	0.8025 $\pm$ 0.0282(10)	0.9570 $\pm$ 0.0075(6)	0.6877 $\pm$ 0.0171(15)	0.7386 $\pm$ 0.0145(4)	0.5947 $\pm$ 0.0167(11)	0.6661 $\pm$ 0.0575(12)
DiffInt	0.8330 $\pm$ 0.0094(8)	0.9721 $\pm$ 0.0050(2)	0.8023 $\pm$ 0.0122(6)	0.7976 $\pm$ 0.0044(2)	0.6540 $\pm$ 0.0045(5)	0.9006 $\pm$ 0.0102(2)
ECOD	0.8249 $\pm$ 0.0099(9)	0.9519 $\pm$ 0.0047(10)	0.8322 $\pm$ 0.0096(3)	0.4919 $\pm$ 0.0126(20)	0.3959 $\pm$ 0.0111(21)	0.5871 $\pm$ 0.0418(17)
GMM	0.8412 $\pm$ 0.0270(7)	0.9448 $\pm$ 0.0075(11)	0.7507 $\pm$ 0.0134(10)	0.7024 $\pm$ 0.0160(9)	0.4967 $\pm$ 0.0096(19)	0.8473 $\pm$ 0.0350(5)
HBOS	0.7367 $\pm$ 0.0502(14)	0.8513 $\pm$ 0.0151(15)	0.6938 $\pm$ 0.0249(14)	0.6047 $\pm$ 0.0649(15)	0.6991 $\pm$ 0.0124(2)	0.5934 $\pm$ 0.0266(16)
IForest	0.9183 $\pm$ 0.0069(2)	0.9524 $\pm$ 0.0094(8)	0.8110 $\pm$ 0.0260(5)	0.6565 $\pm$ 0.0253(11)	0.6003 $\pm$ 0.0210(10)	0.6365 $\pm$ 0.0459(13)
KDE	0.5866 $\pm$ 0.0150(21)	0.9624 $\pm$ 0.0015(5)	0.8378 $\pm$ 0.0079(2)	0.7583 $\pm$ 0.0132(3)	0.5658 $\pm$ 0.0105(15)	0.7064 $\pm$ 0.0385(11)
KPCA	0.7473 $\pm$ 0.0161(12)	0.4696 $\pm$ 0.0601(21)	0.4918 $\pm$ 0.0441(20)	0.5139 $\pm$ 0.0176(19)	0.5017 $\pm$ 0.0162(18)	0.5057 $\pm$ 0.0391(19)
LMDD	0.5926 $\pm$ 0.0166(20)	0.5774 $\pm$ 0.0236(20)	0.7130 $\pm$ 0.0214(13)	0.5351 $\pm$ 0.0282(18)	0.4150 $\pm$ 0.0511(20)	0.4388 $\pm$ 0.0308(21)
LOF	0.7080 $\pm$ 0.0130(17)	0.7194 $\pm$ 0.0416(17)	0.5736 $\pm$ 0.0207(18)	0.6000 $\pm$ 0.0126(16)	0.5340 $\pm$ 0.0131(16)	0.8203 $\pm$ 0.0272(6)
LUNAR	0.8971 $\pm$ 0.0168(3)	0.9631 $\pm$ 0.0038(4)	0.7980 $\pm$ 0.0203(7)	0.8110 $\pm$ 0.0120(1)	0.7870 $\pm$ 0.0066(1)	0.9319 $\pm$ 0.0166(1)
MCD	0.9216 $\pm$ 0.0065(1)	0.8603 $\pm$ 0.0172(14)	0.6641 $\pm$ 0.0230(16)	0.6222 $\pm$ 0.0350(13)	0.6155 $\pm$ 0.0053(8)	0.8111 $\pm$ 0.0230(8)
MO_GAAL	0.5521 $\pm$ 0.0633(22)	0.2338 $\pm$ 0.1069(23)	0.4305 $\pm$ 0.1216(22)	0.4666 $\pm$ 0.0522(22)	0.6447 $\pm$ 0.0239(7)	0.4041 $\pm$ 0.0506(22)
OCSVM	0.5951 $\pm$ 0.0141(19)	0.9521 $\pm$ 0.0034(9)	0.8493 $\pm$ 0.0079(1)	0.5551 $\pm$ 0.0202(17)	0.3788 $\pm$ 0.0105(22)	0.5003 $\pm$ 0.0482(20)
PCA	0.6734 $\pm$ 0.0175(16)	0.7263 $\pm$ 0.2385(16)	0.5786 $\pm$ 0.1266(17)	0.4850 $\pm$ 0.0191(21)	0.3652 $\pm$ 0.0112(23)	0.5287 $\pm$ 0.0401(18)
QMCD	0.7681 $\pm$ 0.0130(11)	0.7117 $\pm$ 0.0402(18)	0.5658 $\pm$ 0.0256(19)	0.6437 $\pm$ 0.0409(12)	0.6834 $\pm$ 0.0094(4)	0.6183 $\pm$ 0.0358(15)
SO_GAAL	0.5461 $\pm$ 0.0616(23)	0.2399 $\pm$ 0.1106(22)	0.4256 $\pm$ 0.1214(23)	0.4652 $\pm$ 0.0450(23)	0.5759 $\pm$ 0.0248(14)	0.4014 $\pm$ 0.0512(23)
Sampling	0.6445 $\pm$ 0.0226(18)	0.8934 $\pm$ 0.0399(13)	0.7324 $\pm$ 0.0492(11)	0.7165 $\pm$ 0.0336(7)	0.6522 $\pm$ 0.0704(6)	0.7229 $\pm$ 0.0388(10)
TCCM	0.8840 $\pm$ 0.1138(4)	0.9775 $\pm$ 0.0094(1)	0.7653 $\pm$ 0.0441(9)	0.7362 $\pm$ 0.0257(6)	0.6044 $\pm$ 0.0186(9)	0.8574 $\pm$ 0.0287(4)
VAE	0.8821 $\pm$ 0.0227(5)	0.9673 $\pm$ 0.0092(3)	0.8153 $\pm$ 0.0201(4)	0.6082 $\pm$ 0.0315(14)	0.5807 $\pm$ 0.0081(12)	0.6259 $\pm$ 0.0420(14)

TABLE XIII: ROC–AUC, medium datasets (2/3).

Model	27_PageBlocks	28_pendigits	30_satellite	31_satimage-2	38_thyroid	40_vowels
ABOD	0.7166 $\pm$ 0.0251(20)	0.6559 $\pm$ 0.0303(18)	0.5619 $\pm$ 0.0148(20)	0.7878 $\pm$ 0.0471(18)	0.9111 $\pm$ 0.0147(16)	0.9440 $\pm$ 0.0194(6)
AutoEncoder	0.9434 $\pm$ 0.0122(4)	0.9805 $\pm$ 0.0078(5)	0.8047 $\pm$ 0.0102(8)	0.9975 $\pm$ 0.0020(7)	0.9816 $\pm$ 0.0080(7)	0.9460 $\pm$ 0.0187(5)
CBLOF	0.8792 $\pm$ 0.0185(14)	0.9760 $\pm$ 0.0054(7)	0.8575 $\pm$ 0.0259(4)	0.9985 $\pm$ 0.0019(3)	0.9354 $\pm$ 0.0105(14)	0.9343 $\pm$ 0.0234(7)
DIF	0.9354 $\pm$ 0.0115(5)	0.9815 $\pm$ 0.0070(4)	0.7886 $\pm$ 0.0133(12)	0.9979 $\pm$ 0.0017(4)	0.9838 $\pm$ 0.0039(6)	0.8136 $\pm$ 0.0444(10)
DiffInt	0.9500 $\pm$ 0.0043(2)	0.9966 $\pm$ 0.0011(2)	0.8416 $\pm$ 0.0046(5)	0.9993 $\pm$ 0.0010(1)	0.9770 $\pm$ 0.0047(8)	0.9808 $\pm$ 0.0068(2)
ECOD	0.9253 $\pm$ 0.0030(7)	0.9329 $\pm$ 0.0094(14)	0.6155 $\pm$ 0.0115(16)	0.9749 $\pm$ 0.0169(17)	0.9859 $\pm$ 0.0025(4)	0.6123 $\pm$ 0.0509(18)
GMM	0.9577 $\pm$ 0.0054(1)	0.8485 $\pm$ 0.0120(16)	0.8028 $\pm$ 0.0065(10)	0.9958 $\pm$ 0.0013(10)	0.9760 $\pm$ 0.0050(10)	0.9480 $\pm$ 0.0177(4)
HBOS	0.8133 $\pm$ 0.0120(16)	0.9377 $\pm$ 0.0095(12)	0.8624 $\pm$ 0.0077(2)	0.9841 $\pm$ 0.0135(15)	0.9877 $\pm$ 0.0039(3)	0.7046 $\pm$ 0.0568(14)
IForest	0.9200 $\pm$ 0.0099(8)	0.9695 $\pm$ 0.0051(8)	0.7967 $\pm$ 0.0151(11)	0.9951 $\pm$ 0.0048(11)	0.9878 $\pm$ 0.0034(2)	0.7874 $\pm$ 0.0307(11)
KDE	0.8868 $\pm$ 0.0099(13)	0.9918 $\pm$ 0.0025(3)	0.7678 $\pm$ 0.0099(13)	0.9977 $\pm$ 0.0031(6)	0.8526 $\pm$ 0.0155(18)	0.7116 $\pm$ 0.0410(13)
KPCA	0.6020 $\pm$ 0.0179(21)	0.5025 $\pm$ 0.0867(20)	0.4778 $\pm$ 0.0156(21)	0.4588 $\pm$ 0.1931(21)	0.3309 $\pm$ 0.0614(23)	0.4887 $\pm$ 0.0458(21)
LMDD	0.7235 $\pm$ 0.0263(19)	0.6073 $\pm$ 0.0723(19)	0.4468 $\pm$ 0.0678(22)	0.5015 $\pm$ 0.1118(20)	0.7074 $\pm$ 0.0709(20)	0.5107 $\pm$ 0.0551(20)
LOF	0.7654 $\pm$ 0.0132(17)	0.4669 $\pm$ 0.0438(21)	0.5634 $\pm$ 0.0141(19)	0.4551 $\pm$ 0.2150(22)	0.9417 $\pm$ 0.0230(13)	0.9497 $\pm$ 0.0148(3)
LUNAR	0.9353 $\pm$ 0.0114(6)	0.9994 $\pm$ 0.0005(1)	0.8803 $\pm$ 0.0073(1)	0.9978 $\pm$ 0.0024(5)	0.9770 $\pm$ 0.0053(8)	0.9887 $\pm$ 0.0052(1)
MCD	0.9195 $\pm$ 0.0075(9)	0.8298 $\pm$ 0.0103(17)	0.8093 $\pm$ 0.0073(7)	0.9960 $\pm$ 0.0010(9)	0.9852 $\pm$ 0.0029(5)	0.7523 $\pm$ 0.0651(12)
MO_GAAL	0.2884 $\pm$ 0.0932(23)	0.2023 $\pm$ 0.0833(22)	0.4177 $\pm$ 0.0347(23)	0.2075 $\pm$ 0.1385(23)	0.7068 $\pm$ 0.0820(21)	0.1081 $\pm$ 0.0740(22)
OCSVM	0.8962 $\pm$ 0.0092(12)	0.9643 $\pm$ 0.0051(9)	0.6537 $\pm$ 0.0126(15)	0.9859 $\pm$ 0.0140(14)	0.8610 $\pm$ 0.0151(17)	0.5997 $\pm$ 0.0462(19)
PCA	0.9061 $\pm$ 0.0075(11)	0.9360 $\pm$ 0.0066(13)	0.5999 $\pm$ 0.0117(18)	0.9820 $\pm$ 0.0155(16)	0.9571 $\pm$ 0.0053(11)	0.6158 $\pm$ 0.0514(17)
QMCD	0.7648 $\pm$ 0.0253(18)	0.9106 $\pm$ 0.0099(15)	0.8605 $\pm$ 0.0072(3)	0.9961 $\pm$ 0.0036(8)	0.8329 $\pm$ 0.0760(19)	0.7020 $\pm$ 0.0460(15)
SO_GAAL	0.2970 $\pm$ 0.0840(22)	0.1966 $\pm$ 0.0956(23)	0.6032 $\pm$ 0.1301(17)	0.7099 $\pm$ 0.2435(19)	0.6946 $\pm$ 0.0803(22)	0.0992 $\pm$ 0.0719(23)
Sampling	0.8629 $\pm$ 0.0352(15)	0.9379 $\pm$ 0.0366(11)	0.8141 $\pm$ 0.0294(6)	0.9946 $\pm$ 0.0072(12)	0.9239 $\pm$ 0.0183(15)	0.8896 $\pm$ 0.0486(9)
TCCM	0.9152 $\pm$ 0.0262(10)	0.9774 $\pm$ 0.0198(6)	0.8029 $\pm$ 0.0111(9)	0.9989 $\pm$ 0.0014(2)	0.9443 $\pm$ 0.0065(12)	0.9339 $\pm$ 0.0172(8)
VAE	0.9477 $\pm$ 0.0099(3)	0.9454 $\pm$ 0.0062(10)	0.7602 $\pm$ 0.0093(14)	0.9872 $\pm$ 0.0081(13)	0.9883 $\pm$ 0.0027(1)	0.6742 $\pm$ 0.0538(16)

TABLE XIV: ROC–AUC, medium datasets (3/3).

Model	41_Waveform	44_Wilt	47_yeast
ABOD	0.6265 $\pm$ 0.0327(15)	0.5544 $\pm$ 0.0226(9)	0.4188 $\pm$ 0.0212(18)
AutoEncoder	0.6848 $\pm$ 0.0565(12)	0.5556 $\pm$ 0.0760(8)	0.4561 $\pm$ 0.0156(9)
CBLOF	0.7546 $\pm$ 0.0522(3)	0.3755 $\pm$ 0.0282(19)	0.4777 $\pm$ 0.0209(6)
DIF	0.7470 $\pm$ 0.0248(5)	0.4179 $\pm$ 0.0614(16)	0.3969 $\pm$ 0.0183(23)
DiffInt	0.7678 $\pm$ 0.0178(1)	0.6789 $\pm$ 0.0272(5)	0.4926 $\pm$ 0.0163(5)
ECOD	0.6140 $\pm$ 0.0352(16)	0.3994 $\pm$ 0.0286(18)	0.4492 $\pm$ 0.0167(12)
GMM	0.5790 $\pm$ 0.0356(17)	0.7295 $\pm$ 0.0295(2)	0.4374 $\pm$ 0.0190(14)
HBOS	0.6973 $\pm$ 0.0316(11)	0.4051 $\pm$ 0.0995(17)	0.4213 $\pm$ 0.0180(17)
IForest	0.7332 $\pm$ 0.0412(8)	0.4687 $\pm$ 0.0343(10)	0.4106 $\pm$ 0.0163(19)
KDE	0.7661 $\pm$ 0.0269(2)	0.3569 $\pm$ 0.0299(21)	0.4034 $\pm$ 0.0169(20)
KPCA	0.5138 $\pm$ 0.0243(21)	0.6498 $\pm$ 0.0243(6)	0.5649 $\pm$ 0.0142(1)
LMDD	0.5173 $\pm$ 0.0232(20)	0.4192 $\pm$ 0.0570(15)	0.4642 $\pm$ 0.0230(7)
LOF	0.7395 $\pm$ 0.0396(7)	0.4265 $\pm$ 0.0151(13)	0.4327 $\pm$ 0.0215(15)
LUNAR	0.7528 $\pm$ 0.0484(4)	0.5917 $\pm$ 0.0676(7)	0.4477 $\pm$ 0.0207(13)
MCD	0.5783 $\pm$ 0.0365(18)	0.8608 $\pm$ 0.0102(1)	0.4541 $\pm$ 0.0228(10)
MO_GAAL	0.3040 $\pm$ 0.0835(23)	0.6966 $\pm$ 0.0301(3)	0.5599 $\pm$ 0.0385(3)
OCSVM	0.5662 $\pm$ 0.0422(19)	0.3574 $\pm$ 0.0313(20)	0.4260 $\pm$ 0.0158(16)
PCA	0.6432 $\pm$ 0.0373(13)	0.4404 $\pm$ 0.0234(12)	0.3982 $\pm$ 0.0167(22)
QMCD	0.7448 $\pm$ 0.0245(6)	0.3565 $\pm$ 0.0351(23)	0.3991 $\pm$ 0.0196(21)
SO_GAAL	0.3466 $\pm$ 0.0815(22)	0.6906 $\pm$ 0.0275(4)	0.5642 $\pm$ 0.0432(2)
Sampling	0.7022 $\pm$ 0.0578(10)	0.4217 $\pm$ 0.0380(14)	0.4602 $\pm$ 0.0513(8)
TCCM	0.6406 $\pm$ 0.0827(14)	0.3567 $\pm$ 0.0623(22)	0.5201 $\pm$ 0.0587(4)
VAE	0.7081 $\pm$ 0.0266(9)	0.4547 $\pm$ 0.0278(11)	0.4500 $\pm$ 0.0162(11)

TABLE XV: ROC–AUC, large datasets (1/2).

Model	1_ALOI	8_celeba	10_cover	11_donors	13_fraud	16_http
ABOD	0.7311 $\pm$ 0.0082(3)	0.4706 $\pm$ 0.0053(19)	0.7391 $\pm$ 0.0072(16)	0.4036 $\pm$ 0.0014(16)	0.8889 $\pm$ 0.0172(17)	0.7510 $\pm$ 0.0017(16)
AutoEncoder	0.5554 $\pm$ 0.0126(6)	0.6988 $\pm$ 0.0311(11)	0.9843 $\pm$ 0.0044(2)	0.9359 $\pm$ 0.0100(3)	0.9592 $\pm$ 0.0070(4)	0.9988 $\pm$ 0.0012(6)
CBLOF	0.5449 $\pm$ 0.0101(11)	0.6790 $\pm$ 0.0629(12)	0.9245 $\pm$ 0.0174(9)	0.9159 $\pm$ 0.0156(6)	0.9535 $\pm$ 0.0052(9)	0.9949 $\pm$ 0.0001(12)
DIF	0.5498 $\pm$ 0.0103(9)	0.6636 $\pm$ 0.0155(14)	0.9787 $\pm$ 0.0040(3)	0.9118 $\pm$ 0.0101(8)	0.9528 $\pm$ 0.0078(10)	0.9935 $\pm$ 0.0002(14)
DiffInt	0.5857 $\pm$ 0.0067(4)	0.7463 $\pm$ 0.0386(8)	0.9949 $\pm$ 0.0015(1)	1.0000 $\pm$ 0.0000(1)	0.9596 $\pm$ 0.0025(3)	0.9998 $\pm$ 0.0022(1)
ECOD	0.5326 $\pm$ 0.0086(16)	0.7664 $\pm$ 0.0030(7)	0.9386 $\pm$ 0.0019(6)	0.9215 $\pm$ 0.0005(5)	0.9525 $\pm$ 0.0057(11)	0.9880 $\pm$ 0.0003(15)
GMM	0.5601 $\pm$ 0.0100(5)	0.8076 $\pm$ 0.0032(3)	0.9505 $\pm$ 0.0012(5)	0.9255 $\pm$ 0.0004(4)	0.9597 $\pm$ 0.0053(2)	0.9994 $\pm$ 0.0000(4)
HBOS	0.5323 $\pm$ 0.0091(17)	0.7685 $\pm$ 0.0031(6)	0.7880 $\pm$ 0.0041(15)	0.7922 $\pm$ 0.0016(13)	0.9551 $\pm$ 0.0060(7)	0.9961 $\pm$ 0.0014(8)
IForest	0.5382 $\pm$ 0.0112(12)	0.7127 $\pm$ 0.0194(9)	0.8456 $\pm$ 0.0235(12)	0.9124 $\pm$ 0.0222(7)	0.9513 $\pm$ 0.0063(12)	0.9938 $\pm$ 0.0029(13)
KDE	0.5367 $\pm$ 0.0103(14)	0.6675 $\pm$ 0.0039(13)	0.8716 $\pm$ 0.0019(10)	-	-	-
KPCA	0.5327 $\pm$ 0.0114(15)	-	-	-	-	-
LMDD	0.5322 $\pm$ 0.0124(18)	-	-	-	-	-
LOF	0.7347 $\pm$ 0.0109(2)	0.4171 $\pm$ 0.0091(20)	0.5547 $\pm$ 0.0107(18)	0.6111 $\pm$ 0.0036(15)	0.4976 $\pm$ 0.0086(18)	0.3929 $\pm$ 0.0052(18)
LUNAR	0.7389 $\pm$ 0.0114(1)	0.6335 $\pm$ 0.0069(15)	-	-	0.9658 $\pm$ 0.0057(1)	0.9953 $\pm$ 0.0037(10)
MCD	0.5262 $\pm$ 0.0088(20)	0.8246 $\pm$ 0.0175(2)	0.7027 $\pm$ 0.0025(17)	0.8202 $\pm$ 0.0993(12)	0.9232 $\pm$ 0.0077(16)	0.9995 $\pm$ 0.0000(3)
MO_GAAL	0.4740 $\pm$ 0.0121(23)	0.2365 $\pm$ 0.0809(21)	-	-	-	-
OCSVM	0.5320 $\pm$ 0.0086(19)	0.7100 $\pm$ 0.0033(10)	0.9251 $\pm$ 0.0020(8)	-	0.9501 $\pm$ 0.0072(13)	-
PCA	0.5485 $\pm$ 0.0107(10)	0.7851 $\pm$ 0.0032(4)	0.9337 $\pm$ 0.0010(7)	0.8240 $\pm$ 0.0011(11)	0.9546 $\pm$ 0.0052(8)	0.9961 $\pm$ 0.0001(8)
QMCD	0.5258 $\pm$ 0.0096(21)	0.5000 $\pm$ 0.0000(17)	0.8215 $\pm$ 0.0030(14)	0.7250 $\pm$ 0.0033(14)	0.9575 $\pm$ 0.0059(5)	0.9968 $\pm$ 0.0002(7)
SO_GAAL	0.4770 $\pm$ 0.0113(22)	0.4803 $\pm$ 0.0248(18)	0.5161 $\pm$ 0.0088(19)	0.3832 $\pm$ 0.1452(17)	0.0751 $\pm$ 0.0216(19)	0.4447 $\pm$ 0.1537(17)
Sampling	0.5376 $\pm$ 0.0108(13)	0.6126 $\pm$ 0.1041(16)	0.8391 $\pm$ 0.0579(13)	0.8793 $\pm$ 0.0257(9)	0.9472 $\pm$ 0.0106(14)	0.9952 $\pm$ 0.0014(11)
TCCM	0.5501 $\pm$ 0.0163(8)	0.8308 $\pm$ 0.0109(1)	0.8705 $\pm$ 0.1019(11)	0.9955 $\pm$ 0.0057(2)	0.9404 $\pm$ 0.0136(15)	0.9997 $\pm$ 0.0007(2)
VAE	0.5541 $\pm$ 0.0084(7)	0.7690 $\pm$ 0.0035(5)	0.9745 $\pm$ 0.0009(4)	0.8428 $\pm$ 0.0458(10)	0.9574 $\pm$ 0.0045(6)	0.9994 $\pm$ 0.0000(4)

TABLE XVI: ROC–AUC, large datasets (2/2).

Model	22_magic	23_mammography	32_shuttle	33_skin	34_smtp
ABOD	0.7930 $\pm$ 0.0069(7)	0.5424 $\pm$ 0.0121(21)	0.6135 $\pm$ 0.0077(20)	0.4064 $\pm$ 0.0026(20)	0.8606 $\pm$ 0.0198(9)
AutoEncoder	0.8211 $\pm$ 0.0111(4)	0.9114 $\pm$ 0.0197(2)	0.9968 $\pm$ 0.0010(3)	0.7975 $\pm$ 0.0685(9)	0.9285 $\pm$ 0.0433(2)
CBLOF	0.8235 $\pm$ 0.0062(3)	0.8238 $\pm$ 0.0188(13)	0.9871 $\pm$ 0.0029(15)	0.9178 $\pm$ 0.0035(3)	0.8446 $\pm$ 0.0747(14)
DIF	0.7822 $\pm$ 0.0136(8)	0.8281 $\pm$ 0.0231(12)	0.9905 $\pm$ 0.0023(9)	0.8759 $\pm$ 0.0087(8)	0.8490 $\pm$ 0.0617(13)
DiffInt	0.8644 $\pm$ 0.0026(1)	0.8803 $\pm$ 0.0097(7)	0.9992 $\pm$ 0.0004(2)	0.9909 $\pm$ 0.0021(1)	0.8961 $\pm$ 0.0151(6)
ECOD	0.6541 $\pm$ 0.0055(19)	0.9115 $\pm$ 0.0163(1)	0.9962 $\pm$ 0.0007(5)	0.5421 $\pm$ 0.0011(18)	0.8868 $\pm$ 0.0419(7)
GMM	0.8040 $\pm$ 0.0036(5)	0.8814 $\pm$ 0.0198(6)	0.9936 $\pm$ 0.0016(8)	0.8885 $\pm$ 0.0004(6)	0.8311 $\pm$ 0.0862(16)
HBOS	0.7301 $\pm$ 0.0060(13)	0.8706 $\pm$ 0.0169(8)	0.9877 $\pm$ 0.0037(13)	0.7722 $\pm$ 0.0011(10)	0.8528 $\pm$ 0.0687(12)
IForest	0.7764 $\pm$ 0.0148(9)	0.8832 $\pm$ 0.0230(4)	0.9965 $\pm$ 0.0008(4)	0.8909 $\pm$ 0.0043(5)	0.9061 $\pm$ 0.0377(5)
KDE	0.7624 $\pm$ 0.0043(11)	0.8581 $\pm$ 0.0153(9)	0.9875 $\pm$ 0.0021(14)	0.7324 $\pm$ 0.0012(12)	0.8026 $\pm$ 0.0518(17)
KPCA	0.5247 $\pm$ 0.0061(20)	0.7860 $\pm$ 0.0135(17)	0.9270 $\pm$ 0.0039(17)	-	-
LMDD	0.5204 $\pm$ 0.0306(21)	0.7440 $\pm$ 0.0772(18)	0.6638 $\pm$ 0.1080(19)	-	0.6184 $\pm$ 0.1197(20)
LOF	0.6790 $\pm$ 0.0044(17)	0.8018 $\pm$ 0.0235(15)	0.5391 $\pm$ 0.0107(21)	0.5472 $\pm$ 0.0032(17)	0.9139 $\pm$ 0.0409(4)
LUNAR	0.8554 $\pm$ 0.0047(2)	0.8817 $\pm$ 0.0209(5)	0.9997 $\pm$ 0.0002(1)	0.9901 $\pm$ 0.0015(2)	0.9280 $\pm$ 0.0266(3)
MCD	0.7380 $\pm$ 0.0043(12)	0.7199 $\pm$ 0.1363(20)	0.9898 $\pm$ 0.0009(11)	0.8844 $\pm$ 0.0004(7)	0.9508 $\pm$ 0.0149(1)
MO_GAAL	0.2031 $\pm$ 0.0117(23)	0.1150 $\pm$ 0.0270(22)	0.0289 $\pm$ 0.0036(23)	-	0.4082 $\pm$ 0.0658(22)
OCSVM	0.7030 $\pm$ 0.0048(16)	0.8486 $\pm$ 0.0127(11)	0.9869 $\pm$ 0.0022(16)	0.7610 $\pm$ 0.0013(11)	0.7982 $\pm$ 0.0538(18)
PCA	0.6666 $\pm$ 0.0062(18)	0.8923 $\pm$ 0.0151(3)	0.9899 $\pm$ 0.0017(10)	0.4149 $\pm$ 0.0026(19)	0.8348 $\pm$ 0.0519(15)
QMCD	0.7157 $\pm$ 0.0053(14)	0.7313 $\pm$ 0.0276(19)	0.7274 $\pm$ 0.0097(18)	0.6219 $\pm$ 0.0010(15)	0.7361 $\pm$ 0.0453(19)
SO_GAAL	0.2044 $\pm$ 0.0145(22)	0.1149 $\pm$ 0.0253(23)	0.0297 $\pm$ 0.0039(22)	0.5628 $\pm$ 0.2584(16)	0.4270 $\pm$ 0.1116(21)
Sampling	0.7934 $\pm$ 0.0237(6)	0.8047 $\pm$ 0.0303(14)	0.9892 $\pm$ 0.0030(12)	0.9165 $\pm$ 0.0304(4)	0.8701 $\pm$ 0.0589(8)
TCCM	0.7748 $\pm$ 0.0216(10)	0.8503 $\pm$ 0.0335(10)	0.9961 $\pm$ 0.0007(6)	0.6414 $\pm$ 0.2798(13)	0.8591 $\pm$ 0.0747(11)
VAE	0.7088 $\pm$ 0.0183(15)	0.7924 $\pm$ 0.0272(16)	0.9948 $\pm$ 0.0012(7)	0.6267 $\pm$ 0.0018(14)	0.8595 $\pm$ 0.0712(10)

TABLE XVII: ROC–AUC, high-dimensional datasets (1/2).

Model	3_backdoor	5_campaign	9_census	17_InternetAds	24_mnist	25_musk
ABOD	0.6392 $\pm$ 0.0091(17)	0.6588 $\pm$ 0.0065(16)	0.4026 $\pm$ 0.0026(9)	0.6577 $\pm$ 0.0328(14)	0.7120 $\pm$ 0.0145(16)	0.2821 $\pm$ 0.0881(20)
AutoEncoder	0.9331 $\pm$ 0.0084(5)	0.8128 $\pm$ 0.0074(1)	-	0.8806 $\pm$ 0.0147(3)	0.9345 $\pm$ 0.0081(1)	1.0000 $\pm$ 0.0000(1)
CBLOF	0.8396 $\pm$ 0.0204(12)	0.6743 $\pm$ 0.0125(15)	-	0.7185 $\pm$ 0.0115(7)	0.8727 $\pm$ 0.0109(11)	1.0000 $\pm$ 0.0000(1)
DIF	0.9311 $\pm$ 0.0067(6)	0.6776 $\pm$ 0.0116(14)	-	0.6346 $\pm$ 0.0217(16)	0.8818 $\pm$ 0.0097(8)	0.9999 $\pm$ 0.0001(11)
DiffInt	0.9613 $\pm$ 0.0043(1)	0.8106 $\pm$ 0.0050(2)	0.6273 $\pm$ 0.0007(4)	0.7036 $\pm$ 0.0053(11)	0.8950 $\pm$ 0.0074(6)	1.0000 $\pm$ 0.0000(1)
ECOD	0.8567 $\pm$ 0.0077(11)	0.7836 $\pm$ 0.0040(8)	-	0.7079 $\pm$ 0.0116(9)	0.7814 $\pm$ 0.0121(14)	0.9984 $\pm$ 0.0008(14)
GMM	0.9276 $\pm$ 0.0083(7)	0.7973 $\pm$ 0.0040(6)	-	0.9123 $\pm$ 0.0089(1)	0.9240 $\pm$ 0.0080(5)	1.0000 $\pm$ 0.0000(1)
HBOS	0.6620 $\pm$ 0.0077(16)	0.8011 $\pm$ 0.0046(5)	-	0.4937 $\pm$ 0.0488(21)	0.6706 $\pm$ 0.0110(17)	1.0000 $\pm$ 0.0000(1)
IForest	0.7612 $\pm$ 0.0352(15)	0.7285 $\pm$ 0.0159(10)	0.6185 $\pm$ 0.0120(5)	0.4246 $\pm$ 0.0516(23)	0.8645 $\pm$ 0.0181(12)	0.9409 $\pm$ 0.0387(16)
KDE	0.9386 $\pm$ 0.0068(4)	0.6999 $\pm$ 0.0065(12)	-	0.8524 $\pm$ 0.0118(6)	0.9336 $\pm$ 0.0067(2)	1.0000 $\pm$ 0.0000(1)
KPCA	-	0.4083 $\pm$ 0.0078(23)	-	0.5284 $\pm$ 0.0289(18)	0.5050 $\pm$ 0.0231(19)	0.5021 $\pm$ 0.0694(17)
LMDD	-	0.4491 $\pm$ 0.0581(20)	-	0.6700 $\pm$ 0.0126(13)	0.3897 $\pm$ 0.0343(23)	0.3448 $\pm$ 0.0445(19)
LOF	0.7669 $\pm$ 0.0131(14)	0.5922 $\pm$ 0.0063(18)	0.5308 $\pm$ 0.0024(6)	0.6397 $\pm$ 0.0372(15)	0.7276 $\pm$ 0.0241(15)	0.3703 $\pm$ 0.0383(18)
LUNAR	0.9539 $\pm$ 0.0064(2)	0.7266 $\pm$ 0.0085(11)	-	0.8835 $\pm$ 0.0133(2)	0.9286 $\pm$ 0.0082(4)	1.0000 $\pm$ 0.0000(1)
MCD	0.8659 $\pm$ 0.0879(10)	0.7961 $\pm$ 0.0082(7)	0.7426 $\pm$ 0.0106(1)	0.5165 $\pm$ 0.0600(19)	0.8790 $\pm$ 0.0245(9)	0.9998 $\pm$ 0.0005(12)
MO_GAAL	0.4091 $\pm$ 0.2470(20)	0.4278 $\pm$ 0.0608(22)	-	0.4862 $\pm$ 0.0435(22)	0.4932 $\pm$ 0.1136(22)	0.2547 $\pm$ 0.1279(22)
OCSVM	0.8668 $\pm$ 0.0070(9)	0.6903 $\pm$ 0.0052(13)	-	0.7079 $\pm$ 0.0117(9)	0.8840 $\pm$ 0.0109(7)	0.9927 $\pm$ 0.0040(15)
PCA	0.5000 $\pm$ 0.0000(19)	0.5000 $\pm$ 0.0000(19)	0.5000 $\pm$ 0.0000(7)	0.6157 $\pm$ 0.0151(17)	0.5000 $\pm$ 0.0000(21)	1.0000 $\pm$ 0.0000(1)
QMCD	0.1021 $\pm$ 0.0065(21)	0.8029 $\pm$ 0.0074(4)	0.0000 $\pm$ 0.0019(10)	0.8591 $\pm$ 0.0185(5)	0.6071 $\pm$ 0.0088(18)	0.0329 $\pm$ 0.0160(23)
SO_GAAL	0.5055 $\pm$ 0.1733(18)	0.4387 $\pm$ 0.0670(21)	0.4850 $\pm$ 0.0353(8)	0.5000 $\pm$ 0.0000(20)	0.5021 $\pm$ 0.1107(20)	0.2675 $\pm$ 0.1256(21)
Sampling	0.8221 $\pm$ 0.0878(13)	0.6508 $\pm$ 0.0233(17)	-	0.7121 $\pm$ 0.0163(8)	0.8177 $\pm$ 0.0664(13)	0.9997 $\pm$ 0.0004(13)
TCCM	0.9526 $\pm$ 0.0111(3)	0.8052 $\pm$ 0.0177(3)	0.6896 $\pm$ 0.0068(3)	0.6984 $\pm$ 0.0109(12)	0.8728 $\pm$ 0.0247(10)	1.0000 $\pm$ 0.0000(1)
VAE	0.9129 $\pm$ 0.0046(8)	0.7800 $\pm$ 0.0039(9)	0.7048 $\pm$ 0.0021(2)	0.8782 $\pm$ 0.0151(4)	0.9332 $\pm$ 0.0083(3)	1.0000 $\pm$ 0.0000(1)

TABLE XVIII: ROC-AUC, high-dimensional datasets (2/2).

Model	26_optdigits	35_SpamBase	36_speech	arrhythmia
ABOD	0.4934 $\pm$ 0.0350(18)	0.4517 $\pm$ 0.0131(20)	0.6062 $\pm$ 0.0881(2)	0.7597 $\pm$ 0.0329(18)
AutoEncoder	0.8707 $\pm$ 0.0148(7)	0.8186 $\pm$ 0.0140(3)	0.4570 $\pm$ 0.0492(13)	0.7890 $\pm$ 0.0327(13)
CBLOF	0.8937 $\pm$ 0.0239(4)	0.6153 $\pm$ 0.0182(13)	0.4515 $\pm$ 0.0473(16)	0.8082 $\pm$ 0.0357(7)
DIF	0.6939 $\pm$ 0.0792(11)	0.5566 $\pm$ 0.0306(17)	0.4397 $\pm$ 0.0622(22)	0.8060 $\pm$ 0.0362(8)
DiffInt	0.9604 $\pm$ 0.0072(3)	0.7041 $\pm$ 0.0072(11)	0.5047 $\pm$ 0.0308(4)	0.8193 $\pm$ 0.0185(5)
ECOD	0.6357 $\pm$ 0.0142(15)	0.7194 $\pm$ 0.0097(9)	0.4511 $\pm$ 0.0486(17)	0.8236 $\pm$ 0.0319(4)
GMM	0.8067 $\pm$ 0.0087(10)	0.7974 $\pm$ 0.0147(7)	0.5013 $\pm$ 0.0483(5)	0.7819 $\pm$ 0.0346(16)
HBOS	0.8763 $\pm$ 0.0137(5)	0.7924 $\pm$ 0.0236(8)	0.4550 $\pm$ 0.0472(14)	0.8287 $\pm$ 0.0309(1)
IForest	0.8421 $\pm$ 0.0192(8)	0.8394 $\pm$ 0.0139(1)	0.4673 $\pm$ 0.0339(10)	0.8247 $\pm$ 0.0345(2)
KDE	0.9969 $\pm$ 0.0009(1)	0.6064 $\pm$ 0.0140(15)	0.6512 $\pm$ 0.0442(1)	0.8150 $\pm$ 0.0298(6)
KPCA	0.4975 $\pm$ 0.0319(17)	0.5190 $\pm$ 0.0172(19)	0.4890 $\pm$ 0.0492(8)	0.4717 $\pm$ 0.0689(20)
LMDD	0.4243 $\pm$ 0.0279(20)	0.5985 $\pm$ 0.0418(16)	0.4750 $\pm$ 0.0355(9)	0.5193 $\pm$ 0.0305(19)
LOF	0.4926 $\pm$ 0.0308(19)	0.4242 $\pm$ 0.0150(21)	0.4666 $\pm$ 0.0494(11)	0.7974 $\pm$ 0.0365(10)
LUNAR	0.9966 $\pm$ 0.0015(2)	0.8229 $\pm$ 0.0234(2)	0.5499 $\pm$ 0.0944(3)	0.8239 $\pm$ 0.0310(3)
MCD	0.6460 $\pm$ 0.0212(13)	0.8001 $\pm$ 0.0260(6)	0.4956 $\pm$ 0.0460(6)	0.7812 $\pm$ 0.0400(17)
MO_GAAL	0.3687 $\pm$ 0.1581(21)	0.3253 $\pm$ 0.0590(22)	0.4483 $\pm$ 0.0608(19)	0.3027 $\pm$ 0.0591(22)
OCSVM	0.6456 $\pm$ 0.0172(14)	0.6178 $\pm$ 0.0142(12)	0.4459 $\pm$ 0.0481(20)	0.8005 $\pm$ 0.0279(9)
PCA	0.5000 $\pm$ 0.0000(16)	0.5500 $\pm$ 0.0125(18)	0.4510 $\pm$ 0.0486(18)	0.7858 $\pm$ 0.0339(15)
QMCD	0.1303 $\pm$ 0.0179(23)	0.7141 $\pm$ 0.0237(10)	0.4217 $\pm$ 0.0446(23)	0.3366 $\pm$ 0.0624(21)
SO_GAAL	0.3658 $\pm$ 0.1716(22)	0.3198 $\pm$ 0.0548(23)	0.4457 $\pm$ 0.0656(21)	0.3005 $\pm$ 0.0800(23)
Sampling	0.8181 $\pm$ 0.0723(9)	0.6115 $\pm$ 0.0168(14)	0.4644 $\pm$ 0.0582(12)	0.7964 $\pm$ 0.0368(11)
TCCM	0.8725 $\pm$ 0.1032(6)	0.8092 $\pm$ 0.0214(5)	0.4899 $\pm$ 0.0480(7)	0.7886 $\pm$ 0.0370(14)
VAE	0.6583 $\pm$ 0.0277(12)	0.8118 $\pm$ 0.0146(4)	0.4516 $\pm$ 0.0489(15)	0.7926 $\pm$ 0.0397(12)

TABLE XIX: AUPRC, small datasets (1/2).

Model	4_breastw	14_glass	15_Hepatitis	18_Ionosphere	21_Lymphography	29_Pima
ABOD	0.2877 $\pm$ 0.0169(20)	0.2252 $\pm$ 0.1340(8)	0.3422 $\pm$ 0.1897(19)	0.9189 $\pm$ 0.0188(9)	0.6225 $\pm$ 0.2143(18)	0.4735 $\pm$ 0.0308(16)
AutoEncoder	0.9664 $\pm$ 0.0184(13)	0.1502 $\pm$ 0.1025(16)	0.6348 $\pm$ 0.1385(2)	0.9416 $\pm$ 0.0131(5)	0.8167 $\pm$ 0.1820(15)	0.5387 $\pm$ 0.0181(7)
CBLOF	0.9892 $\pm$ 0.0043(4)	0.2286 $\pm$ 0.1182(7)	0.5388 $\pm$ 0.1740(4)	0.9201 $\pm$ 0.0374(8)	0.9299 $\pm$ 0.1435(10)	0.5244 $\pm$ 0.0279(10)
DIF	0.6048 $\pm$ 0.0525(17)	0.2519 $\pm$ 0.1434(4)	0.4093 $\pm$ 0.1430(17)	0.9367 $\pm$ 0.0108(6)	0.4838 $\pm$ 0.2684(19)	0.4144 $\pm$ 0.0207(20)
DiffInt	0.9919 $\pm$ 0.0032(1)	0.3294 $\pm$ 0.0800(1)	0.5211 $\pm$ 0.0667(8)	0.9574 $\pm$ 0.0073(2)	0.9951 $\pm$ 0.0201(2)	0.5668 $\pm$ 0.0149(3)
ECOD	0.9869 $\pm$ 0.0053(7)	0.1812 $\pm$ 0.1485(11)	0.4933 $\pm$ 0.0920(10)	0.6874 $\pm$ 0.0444(17)	0.9867 $\pm$ 0.0292(4)	0.4992 $\pm$ 0.0179(13)
GMM	0.9693 $\pm$ 0.0139(12)	0.2004 $\pm$ 0.1567(10)	0.5098 $\pm$ 0.1610(9)	0.9569 $\pm$ 0.0099(3)	0.7933 $\pm$ 0.1614(16)	0.5418 $\pm$ 0.0200(6)
HBOS	0.9799 $\pm$ 0.0088(9)	0.1794 $\pm$ 0.1047(12)	0.5721 $\pm$ 0.0927(3)	0.5333 $\pm$ 0.0867(19)	0.9637 $\pm$ 0.0659(8)	0.5991 $\pm$ 0.0253(2)
IForest	0.9890 $\pm$ 0.0053(5)	0.1543 $\pm$ 0.0638(14)	0.5316 $\pm$ 0.1013(5)	0.8529 $\pm$ 0.0338(13)	0.9900 $\pm$ 0.0211(3)	0.5572 $\pm$ 0.0257(4)
KDE	0.9904 $\pm$ 0.0041(2)	0.1668 $\pm$ 0.1048(13)	0.4333 $\pm$ 0.0932(15)	0.9323 $\pm$ 0.0105(7)	0.9833 $\pm$ 0.0351(5)	0.5295 $\pm$ 0.0295(9)
KPCA	0.2457 $\pm$ 0.0273(21)	0.0525 $\pm$ 0.0186(23)	0.2266 $\pm$ 0.1709(21)	0.3667 $\pm$ 0.0443(21)	0.1209 $\pm$ 0.1796(20)	0.3502 $\pm$ 0.0284(21)
LMDD	0.7921 $\pm$ 0.1227(16)	0.0951 $\pm$ 0.0897(22)	0.2530 $\pm$ 0.1153(20)	0.5672 $\pm$ 0.0604(18)	0.7597 $\pm$ 0.1644(17)	0.4895 $\pm$ 0.0585(14)
LOF	0.3152 $\pm$ 0.0156(19)	0.2320 $\pm$ 0.1363(6)	0.4741 $\pm$ 0.1082(14)	0.8282 $\pm$ 0.0611(14)	0.9833 $\pm$ 0.0351(5)	0.4409 $\pm$ 0.0227(18)
LUNAR	0.9798 $\pm$ 0.0101(10)	0.2542 $\pm$ 0.1299(2)	0.4797 $\pm$ 0.1632(13)	0.9677 $\pm$ 0.0119(1)	0.8629 $\pm$ 0.1506(13)	0.5502 $\pm$ 0.0212(5)
MCD	0.9630 $\pm$ 0.0222(14)	0.1475 $\pm$ 0.0490(17)	0.4847 $\pm$ 0.1319(12)	0.9534 $\pm$ 0.0115(4)	0.8314 $\pm$ 0.1602(14)	0.5054 $\pm$ 0.0295(11)
MO_GAAL	0.2116 $\pm$ 0.0082(22)	0.1409 $\pm$ 0.1880(19)	0.2080 $\pm$ 0.1225(22)	0.2876 $\pm$ 0.0459(22)	0.0339 $\pm$ 0.0072(22)	0.2784 $\pm$ 0.0480(22)
OCSVM	0.9904 $\pm$ 0.0041(2)	0.1249 $\pm$ 0.1108(21)	0.5231 $\pm$ 0.1038(7)	0.8001 $\pm$ 0.0329(15)	0.9833 $\pm$ 0.0351(5)	0.5028 $\pm$ 0.0288(12)
PCA	0.9507 $\pm$ 0.0178(15)	0.1454 $\pm$ 0.1055(18)	0.4853 $\pm$ 0.1374(11)	0.7236 $\pm$ 0.0393(16)	0.9333 $\pm$ 0.1291(9)	0.4678 $\pm$ 0.0199(17)
QMCD	0.5168 $\pm$ 0.1374(18)	0.2025 $\pm$ 0.0687(9)	0.3999 $\pm$ 0.1074(18)	0.3946 $\pm$ 0.0602(20)	0.0378 $\pm$ 0.0084(21)	0.6099 $\pm$ 0.0347(1)
SO_GAAL	0.2116 $\pm$ 0.0082(22)	0.1298 $\pm$ 0.1725(20)	0.1719 $\pm$ 0.0960(23)	0.2702 $\pm$ 0.0379(23)	0.0338 $\pm$ 0.0071(23)	0.2784 $\pm$ 0.0450(22)
Sampling	0.9878 $\pm$ 0.0061(6)	0.2452 $\pm$ 0.1173(5)	0.4322 $\pm$ 0.1248(16)	0.9093 $\pm$ 0.0388(10)	0.8901 $\pm$ 0.1443(11)	0.5312 $\pm$ 0.0416(8)
TCCM	0.9718 $\pm$ 0.0148(11)	0.2536 $\pm$ 0.1251(3)	0.5267 $\pm$ 0.0975(6)	0.9071 $\pm$ 0.0227(11)	1.0000 $\pm$ 0.0000(1)	0.4202 $\pm$ 0.0638(19)
VAE	0.9806 $\pm$ 0.0108(8)	0.1503 $\pm$ 0.0995(15)	0.6372 $\pm$ 0.1153(1)	0.8817 $\pm$ 0.0283(12)	0.8796 $\pm$ 0.1430(12)	0.4876 $\pm$ 0.0228(15)

TABLE XX: AUPRC, small datasets (2/2).

Model	37_Stamps	39_vertebral	42_WBC	43_WDBC	45_wine	46_WPBC
ABOD	0.2120 $\pm$ 0.0573(19)	0.1222 $\pm$ 0.0268(17)	0.3421 $\pm$ 0.1683(18)	0.2573 $\pm$ 0.1460(18)	0.1717 $\pm$ 0.0927(20)	0.2141 $\pm$ 0.0518(22)
AutoEncoder	0.4350 $\pm$ 0.1051(13)	0.1265 $\pm$ 0.0253(14)	0.8106 $\pm$ 0.2272(13)	0.7782 $\pm$ 0.2134(3)	0.5529 $\pm$ 0.2822(6)	0.2347 $\pm$ 0.0631(12)
CBLOF	0.5528 $\pm$ 0.1300(2)	0.1314 $\pm$ 0.0278(9)	0.9480 $\pm$ 0.0504(5)	0.6179 $\pm$ 0.2237(13)	0.6318 $\pm$ 0.2372(2)	0.2413 $\pm$ 0.0610(6)
DIF	0.5181 $\pm$ 0.0937(4)	0.1855 $\pm$ 0.0487(3)	0.1239 $\pm$ 0.0324(19)	0.0662 $\pm$ 0.0339(20)	0.2498 $\pm$ 0.1465(18)	0.2223 $\pm$ 0.0594(19)
DiffInt	0.6095 $\pm$ 0.0647(1)	0.1548 $\pm$ 0.0192(5)	0.9658 $\pm$ 0.0466(1)	0.5729 $\pm$ 0.1161(16)	0.4625 $\pm$ 0.1262(11)	0.2834 $\pm$ 0.0336(1)
ECOD	0.4575 $\pm$ 0.1030(11)	0.1310 $\pm$ 0.0239(10)	0.9174 $\pm$ 0.0831(8)	0.6875 $\pm$ 0.1531(6)	0.3033 $\pm$ 0.0885(16)	0.2158 $\pm$ 0.0474(21)
GMM	0.4618 $\pm$ 0.0950(10)	0.1291 $\pm$ 0.0219(11)	0.6966 $\pm$ 0.1988(17)	0.6851 $\pm$ 0.2442(8)	0.6901 $\pm$ 0.2396(1)	0.2392 $\pm$ 0.0704(8)
HBOS	0.4570 $\pm$ 0.0663(12)	0.1110 $\pm$ 0.0243(21)	0.8305 $\pm$ 0.1530(12)	0.7374 $\pm$ 0.1368(4)	0.5273 $\pm$ 0.2059(9)	0.2355 $\pm$ 0.0603(11)
IForest	0.5178 $\pm$ 0.1048(5)	0.1224 $\pm$ 0.0263(16)	0.9201 $\pm$ 0.0784(7)	0.6873 $\pm$ 0.1773(7)	0.5717 $\pm$ 0.1853(5)	0.2344 $\pm$ 0.0597(13)
KDE	0.4759 $\pm$ 0.0940(8)	0.1144 $\pm$ 0.0241(19)	0.9502 $\pm$ 0.0568(3)	0.6224 $\pm$ 0.1666(12)	0.5357 $\pm$ 0.1809(7)	0.2503 $\pm$ 0.0598(5)
KPCA	0.0939 $\pm$ 0.0238(21)	0.1707 $\pm$ 0.0392(4)	0.1196 $\pm$ 0.1020(21)	0.0208 $\pm$ 0.0054(21)	0.1054 $\pm$ 0.0808(21)	0.2364 $\pm$ 0.0843(10)
LMDD	0.1085 $\pm$ 0.0399(20)	0.1233 $\pm$ 0.0291(15)	0.9302 $\pm$ 0.0867(6)	0.7950 $\pm$ 0.1495(1)	0.2445 $\pm$ 0.1397(19)	0.2394 $\pm$ 0.0452(7)
LOF	0.2330 $\pm$ 0.0598(18)	0.1073 $\pm$ 0.0211(23)	0.7069 $\pm$ 0.2313(16)	0.6436 $\pm$ 0.1994(10)	0.4505 $\pm$ 0.1887(12)	0.2377 $\pm$ 0.0612(9)
LUNAR	0.5381 $\pm$ 0.1321(3)	0.1283 $\pm$ 0.0321(13)	0.7695 $\pm$ 0.2384(15)	0.7913 $\pm$ 0.1626(2)	0.6064 $\pm$ 0.2519(3)	0.2513 $\pm$ 0.0619(4)
MCD	0.2714 $\pm$ 0.0547(17)	0.1342 $\pm$ 0.0274(8)	0.7814 $\pm$ 0.2436(14)	0.4138 $\pm$ 0.1816(17)	0.5772 $\pm$ 0.2406(4)	0.2583 $\pm$ 0.0910(3)
MO_GAAL	0.0723 $\pm$ 0.0240(22)	0.2425 $\pm$ 0.0945(1)	0.0282 $\pm$ 0.0047(22)	0.0158 $\pm$ 0.0044(22)	0.0440 $\pm$ 0.0147(22)	0.2211 $\pm$ 0.0706(20)
OCSVM	0.4249 $\pm$ 0.1001(14)	0.1202 $\pm$ 0.0253(18)	0.9502 $\pm$ 0.0568(3)	0.6429 $\pm$ 0.1641(11)	0.3484 $\pm$ 0.1777(14)	0.2333 $\pm$ 0.0561(14)
PCA	0.4083 $\pm$ 0.0799(15)	0.1093 $\pm$ 0.0161(22)	0.9009 $\pm$ 0.0873(9)	0.6025 $\pm$ 0.1176(14)	0.3114 $\pm$ 0.1194(15)	0.2276 $\pm$ 0.0529(18)
QMCD	0.4661 $\pm$ 0.0532(9)	0.1144 $\pm$ 0.0252(19)	0.1227 $\pm$ 0.0592(20)	0.2214 $\pm$ 0.1521(19)	0.2567 $\pm$ 0.1461(17)	0.2330 $\pm$ 0.0558(15)
SO_GAAL	0.0658 $\pm$ 0.0152(23)	0.2141 $\pm$ 0.0595(2)	0.0282 $\pm$ 0.0047(22)	0.0158 $\pm$ 0.0044(22)	0.0438 $\pm$ 0.0149(23)	0.2100 $\pm$ 0.0529(23)
Sampling	0.4803 $\pm$ 0.1574(7)	0.1291 $\pm$ 0.0358(11)	0.8885 $\pm$ 0.2170(10)	0.5865 $\pm$ 0.1612(15)	0.5280 $\pm$ 0.1743(8)	0.2612 $\pm$ 0.0817(2)
TCCM	0.3988 $\pm$ 0.0925(16)	0.1363 $\pm$ 0.0268(7)	0.9610 $\pm$ 0.0576(2)	0.6813 $\pm$ 0.1950(9)	0.5198 $\pm$ 0.2062(10)	0.2295 $\pm$ 0.0547(16)
VAE	0.4855 $\pm$ 0.0914(6)	0.1434 $\pm$ 0.0283(6)	0.8563 $\pm$ 0.1843(11)	0.6896 $\pm$ 0.1968(5)	0.4389 $\pm$ 0.1942(13)	0.2283 $\pm$ 0.0544(17)

TABLE XXI: AUPRC, medium datasets (1/3).

Model	2_anthyroid	6_cardio	7_Cardiotocography	12_fault	19_landsat	20_letter
ABOD	0.1685 $\pm$ 0.0118(17)	0.1865 $\pm$ 0.0217(19)	0.2591 $\pm$ 0.0182(20)	0.5010 $\pm$ 0.0160(10)	0.2219 $\pm$ 0.0116(18)	0.3120 $\pm$ 0.0422(6)
AutoEncoder	0.5003 $\pm$ 0.0592(4)	0.6195 $\pm$ 0.0650(12)	0.4920 $\pm$ 0.0476(13)	0.5937 $\pm$ 0.0420(5)	0.2604 $\pm$ 0.0116(10)	0.2659 $\pm$ 0.0448(7)
CBLOF	0.1605 $\pm$ 0.0144(18)	0.7159 $\pm$ 0.0405(7)	0.5803 $\pm$ 0.0271(7)	0.5318 $\pm$ 0.0543(9)	0.3355 $\pm$ 0.0152(4)	0.1951 $\pm$ 0.0339(9)
DIF	0.4211 $\pm$ 0.0358(8)	0.7767 $\pm$ 0.0339(4)	0.5001 $\pm$ 0.0246(12)	0.5894 $\pm$ 0.0342(6)	0.2932 $\pm$ 0.0103(7)	0.1555 $\pm$ 0.0476(10)
DiffInt	0.3652 $\pm$ 0.0133(10)	0.8114 $\pm$ 0.0213(2)	0.6271 $\pm$ 0.0166(1)	0.6639 $\pm$ 0.0128(2)	0.3688 $\pm$ 0.0088(2)	0.3954 $\pm$ 0.0306(2)
ECOD	0.3479 $\pm$ 0.0222(11)	0.6705 $\pm$ 0.0231(11)	0.5820 $\pm$ 0.0238(6)	0.3467 $\pm$ 0.0165(22)	0.1769 $\pm$ 0.0043(21)	0.0899 $\pm$ 0.0180(16)
GMM	0.4365 $\pm$ 0.0291(7)	0.6968 $\pm$ 0.0376(8)	0.5269 $\pm$ 0.0250(9)	0.5459 $\pm$ 0.0311(7)	0.2060 $\pm$ 0.0064(20)	0.3384 $\pm$ 0.0526(3)
HBOS	0.3934 $\pm$ 0.0471(9)	0.5475 $\pm$ 0.0316(14)	0.4559 $\pm$ 0.0241(14)	0.4442 $\pm$ 0.0669(15)	0.3580 $\pm$ 0.0178(3)	0.0867 $\pm$ 0.0190(18)
IForest	0.4849 $\pm$ 0.0345(5)	0.6925 $\pm$ 0.0521(9)	0.5648 $\pm$ 0.0281(8)	0.4958 $\pm$ 0.0382(11)	0.2668 $\pm$ 0.0149(9)	0.1011 $\pm$ 0.0206(15)
KDE	0.1274 $\pm$ 0.0129(20)	0.7267 $\pm$ 0.0282(6)	0.6152 $\pm$ 0.0210(3)	0.6196 $\pm$ 0.0279(4)	0.2385 $\pm$ 0.0055(14)	0.1506 $\pm$ 0.0325(11)
KPCA	0.2036 $\pm$ 0.0217(13)	0.0928 $\pm$ 0.0123(22)	0.2136 $\pm$ 0.0213(21)	0.3633 $\pm$ 0.0188(19)	0.2150 $\pm$ 0.0096(19)	0.0714 $\pm$ 0.0089(20)
LMDD	0.1165 $\pm$ 0.0131(21)	0.1569 $\pm$ 0.0189(20)	0.4269 $\pm$ 0.0372(15)	0.3829 $\pm$ 0.0243(18)	0.2274 $\pm$ 0.0884(17)	0.0638 $\pm$ 0.0079(21)
LOF	0.1901 $\pm$ 0.0185(15)	0.2026 $\pm$ 0.0235(18)	0.2883 $\pm$ 0.0292(18)	0.4002 $\pm$ 0.0206(17)	0.2457 $\pm$ 0.0095(13)	0.3357 $\pm$ 0.0444(4)
LUNAR	0.4683 $\pm$ 0.0401(6)	0.7791 $\pm$ 0.0356(3)	0.6128 $\pm$ 0.0307(4)	0.6845 $\pm$ 0.0358(1)	0.5202 $\pm$ 0.0228(1)	0.5013 $\pm$ 0.0652(1)
MCD	0.5152 $\pm$ 0.0299(2)	0.3805 $\pm$ 0.0375(15)	0.3595 $\pm$ 0.0297(16)	0.4347 $\pm$ 0.0468(16)	0.2593 $\pm$ 0.0042(11)	0.1998 $\pm$ 0.0415(8)
MO_GAAL	0.1013 $\pm$ 0.0212(22)	0.0918 $\pm$ 0.0954(23)	0.2120 $\pm$ 0.0732(22)	0.3535 $\pm$ 0.0403(20)	0.3300 $\pm$ 0.0331(5)	0.0577 $\pm$ 0.0097(22)
OCSVM	0.1313 $\pm$ 0.0133(19)	0.6923 $\pm$ 0.0300(10)	0.6186 $\pm$ 0.0240(2)	0.4703 $\pm$ 0.0216(13)	0.1712 $\pm$ 0.0046(22)	0.0807 $\pm$ 0.0169(19)
PCA	0.2015 $\pm$ 0.0113(14)	0.3667 $\pm$ 0.2827(16)	0.2990 $\pm$ 0.1234(17)	0.3415 $\pm$ 0.0173(23)	0.1635 $\pm$ 0.0041(23)	0.0878 $\pm$ 0.0205(17)
QMCD	0.2606 $\pm$ 0.0275(12)	0.3538 $\pm$ 0.0571(17)	0.2713 $\pm$ 0.0218(19)	0.4705 $\pm$ 0.0492(12)	0.3152 $\pm$ 0.0113(6)	0.1023 $\pm$ 0.0231(14)
SO_GAAL	0.0981 $\pm$ 0.0196(23)	0.0934 $\pm$ 0.0961(21)	0.2083 $\pm$ 0.0727(23)	0.3521 $\pm$ 0.0374(21)	0.2349 $\pm$ 0.0285(16)	0.0559 $\pm$ 0.0080(23)
Sampling	0.1741 $\pm$ 0.0209(16)	0.6043 $\pm$ 0.0843(13)	0.5187 $\pm$ 0.0438(11)	0.5323 $\pm$ 0.0531(8)	0.2839 $\pm$ 0.0537(8)	0.1489 $\pm$ 0.0309(12)
TCCM	0.5279 $\pm$ 0.0935(1)	0.8352 $\pm$ 0.0315(1)	0.5254 $\pm$ 0.0509(10)	0.6234 $\pm$ 0.0299(3)	0.2533 $\pm$ 0.0126(12)	0.3177 $\pm$ 0.0507(5)
VAE	0.5096 $\pm$ 0.0311(3)	0.7723 $\pm$ 0.0403(5)	0.6065 $\pm$ 0.0273(5)	0.4672 $\pm$ 0.0354(14)	0.2372 $\pm$ 0.0057(15)	0.1301 $\pm$ 0.0299(13)

TABLE XXII: AUPRC, medium datasets (2/3).

Model	27_PageBlocks	28_pendigits	30_satellite	31_satimage-2	38_thyroid	40_vowels
ABOD	0.3540 $\pm$ 0.0282(18)	0.0601 $\pm$ 0.0106(18)	0.3867 $\pm$ 0.0150(20)	0.1329 $\pm$ 0.0474(18)	0.1251 $\pm$ 0.0139(21)	0.4213 $\pm$ 0.1150(7)
AutoEncoder	0.7179 $\pm$ 0.0330(4)	0.4177 $\pm$ 0.0942(7)	0.7908 $\pm$ 0.0112(6)	0.8612 $\pm$ 0.0970(12)	0.6788 $\pm$ 0.0687(5)	0.5336 $\pm$ 0.0721(4)
CBLOF	0.5343 $\pm$ 0.0463(12)	0.4306 $\pm$ 0.1145(6)	0.7996 $\pm$ 0.0198(5)	0.9727 $\pm$ 0.0222(2)	0.2328 $\pm$ 0.0654(16)	0.3702 $\pm$ 0.1387(8)
DIF	0.6854 $\pm$ 0.0315(6)	0.5697 $\pm$ 0.1402(5)	0.7803 $\pm$ 0.0134(7)	0.9036 $\pm$ 0.0684(9)	0.7491 $\pm$ 0.0391(1)	0.2130 $\pm$ 0.0693(10)
DiffInt	0.7673 $\pm$ 0.0121(1)	0.8617 $\pm$ 0.0325(2)	0.8168 $\pm$ 0.0060(2)	0.9727 $\pm$ 0.0117(2)	0.5704 $\pm$ 0.0412(10)	0.7692 $\pm$ 0.0502(2)
ECOD	0.5734 $\pm$ 0.0258(9)	0.3003 $\pm$ 0.0431(11)	0.5882 $\pm$ 0.0081(17)	0.7695 $\pm$ 0.0614(15)	0.6398 $\pm$ 0.0425(8)	0.1090 $\pm$ 0.0263(16)
GMM	0.7209 $\pm$ 0.0241(3)	0.0879 $\pm$ 0.0160(16)	0.7768 $\pm$ 0.0111(8)	0.7337 $\pm$ 0.1016(16)	0.6474 $\pm$ 0.0461(7)	0.6293 $\pm$ 0.0983(3)
HBOS	0.3019 $\pm$ 0.0160(19)	0.2869 $\pm$ 0.0357(12)	0.8109 $\pm$ 0.0092(4)	0.8270 $\pm$ 0.0647(13)	0.7392 $\pm$ 0.0772(3)	0.1337 $\pm$ 0.0498(13)
IForest	0.5398 $\pm$ 0.0328(11)	0.3881 $\pm$ 0.0638(8)	0.7662 $\pm$ 0.0140(10)	0.9210 $\pm$ 0.0427(7)	0.6635 $\pm$ 0.0942(6)	0.1938 $\pm$ 0.0504(11)
KDE	0.4864 $\pm$ 0.0316(15)	0.7551 $\pm$ 0.0752(3)	0.7479 $\pm$ 0.0115(12)	0.9771 $\pm$ 0.0245(1)	0.1606 $\pm$ 0.0532(20)	0.1787 $\pm$ 0.0573(12)
KPCA	0.1674 $\pm$ 0.0129(21)	0.0228 $\pm$ 0.0061(21)	0.3056 $\pm$ 0.0142(22)	0.0628 $\pm$ 0.1632(20)	0.0273 $\pm$ 0.0118(23)	0.0378 $\pm$ 0.0092(21)
LMDD	0.4825 $\pm$ 0.0221(16)	0.0283 $\pm$ 0.0061(20)	0.3157 $\pm$ 0.0782(21)	0.0212 $\pm$ 0.0224(22)	0.0696 $\pm$ 0.0161(22)	0.0498 $\pm$ 0.0255(20)
LOF	0.4083 $\pm$ 0.0256(17)	0.0376 $\pm$ 0.0119(19)	0.3932 $\pm$ 0.0145(19)	0.0305 $\pm$ 0.0175(21)	0.2889 $\pm$ 0.0935(13)	0.4563 $\pm$ 0.0951(6)
LUNAR	0.7430 $\pm$ 0.0304(2)	0.9659 $\pm$ 0.0331(1)	0.8405 $\pm$ 0.0110(1)	0.9656 $\pm$ 0.0268(5)	0.6089 $\pm$ 0.0850(9)	0.8212 $\pm$ 0.0922(1)
MCD	0.6209 $\pm$ 0.0226(8)	0.0722 $\pm$ 0.0102(17)	0.7754 $\pm$ 0.0136(9)	0.7194 $\pm$ 0.0769(17)	0.7096 $\pm$ 0.0493(4)	0.0977 $\pm$ 0.0686(17)
MO_GAAL	0.1085 $\pm$ 0.0568(23)	0.0134 $\pm$ 0.0016(22)	0.2918 $\pm$ 0.0194(23)	0.0080 $\pm$ 0.0029(23)	0.2146 $\pm$ 0.1110(17)	0.0191 $\pm$ 0.0025(22)
OCSVM	0.5083 $\pm$ 0.0317(14)	0.3706 $\pm$ 0.0533(9)	0.6837 $\pm$ 0.0107(15)	0.9066 $\pm$ 0.0385(8)	0.1701 $\pm$ 0.0539(19)	0.0954 $\pm$ 0.0293(18)
PCA	0.5327 $\pm$ 0.0308(13)	0.2323 $\pm$ 0.0378(14)	0.6038 $\pm$ 0.0099(16)	0.8750 $\pm$ 0.0433(11)	0.3753 $\pm$ 0.0389(11)	0.0816 $\pm$ 0.0254(19)
QMCD	0.2177 $\pm$ 0.0127(20)	0.1684 $\pm$ 0.0255(15)	0.8145 $\pm$ 0.0096(3)	0.9346 $\pm$ 0.0343(6)	0.3499 $\pm$ 0.0531(12)	0.1307 $\pm$ 0.0329(14)
SO_GAAL	0.1109 $\pm$ 0.0540(22)	0.0134 $\pm$ 0.0013(22)	0.4639 $\pm$ 0.1478(18)	0.1315 $\pm$ 0.2360(19)	0.2050 $\pm$ 0.0953(18)	0.0191 $\pm$ 0.0027(22)
Sampling	0.5436 $\pm$ 0.0792(10)	0.3038 $\pm$ 0.2087(10)	0.7445 $\pm$ 0.0386(13)	0.8896 $\pm$ 0.1368(10)	0.2644 $\pm$ 0.0744(15)	0.3241 $\pm$ 0.0924(9)
TCCM	0.7055 $\pm$ 0.0705(5)	0.6903 $\pm$ 0.1344(4)	0.7586 $\pm$ 0.0199(11)	0.9695 $\pm$ 0.0281(4)	0.2650 $\pm$ 0.0576(14)	0.4667 $\pm$ 0.1105(5)
VAE	0.6740 $\pm$ 0.0416(7)	0.2768 $\pm$ 0.0448(13)	0.7325 $\pm$ 0.0113(14)	0.8211 $\pm$ 0.0622(14)	0.7433 $\pm$ 0.0486(2)	0.1170 $\pm$ 0.0567(15)

TABLE XXIII: AUPRC, medium datasets (3/3).

Model	41_Waveform	44_Wilt	47_yeast
ABOD	0.0432 $\pm$ 0.0084(15)	0.0548 $\pm$ 0.0024(9)	0.3089 $\pm$ 0.0129(19)
AutoEncoder	0.0787 $\pm$ 0.0299(7)	0.0562 $\pm$ 0.0139(8)	0.3197 $\pm$ 0.0182(13)
CBLOF	0.2284 $\pm$ 0.0701(1)	0.0387 $\pm$ 0.0030(19)	0.3223 $\pm$ 0.0202(11)
DIF	0.0873 $\pm$ 0.0246(6)	0.0419 $\pm$ 0.0059(18)	0.2961 $\pm$ 0.0133(23)
DiffInt	0.0921 $\pm$ 0.0117(5)	0.0798 $\pm$ 0.0060(5)	0.3534 $\pm$ 0.0129(5)
ECOD	0.0431 $\pm$ 0.0086(16)	0.0428 $\pm$ 0.0048(15)	0.3393 $\pm$ 0.0157(6)
GMM	0.0426 $\pm$ 0.0086(18)	0.0918 $\pm$ 0.0082(3)	0.3096 $\pm$ 0.0171(16)
HBOS	0.0508 $\pm$ 0.0073(13)	0.0444 $\pm$ 0.0094(13)	0.3334 $\pm$ 0.0165(8)
IForest	0.0626 $\pm$ 0.0121(9)	0.0453 $\pm$ 0.0033(12)	0.3096 $\pm$ 0.0167(16)
KDE	0.0696 $\pm$ 0.0097(8)	0.0385 $\pm$ 0.0030(21)	0.2974 $\pm$ 0.0143(21)
KPCA	0.0320 $\pm$ 0.0048(21)	0.0717 $\pm$ 0.0053(6)	0.3807 $\pm$ 0.0174(1)
LMDD	0.0323 $\pm$ 0.0044(20)	0.0494 $\pm$ 0.0070(10)	0.3337 $\pm$ 0.0185(7)
LOF	0.1167 $\pm$ 0.0347(4)	0.0426 $\pm$ 0.0029(16)	0.3091 $\pm$ 0.0220(18)
LUNAR	0.2161 $\pm$ 0.0711(2)	0.0612 $\pm$ 0.0128(7)	0.3226 $\pm$ 0.0206(10)
MCD	0.0427 $\pm$ 0.0091(17)	0.1547 $\pm$ 0.0106(1)	0.3211 $\pm$ 0.0242(12)
MO_GAAL	0.0212 $\pm$ 0.0064(23)	0.0928 $\pm$ 0.0121(2)	0.3647 $\pm$ 0.0406(3)
OCSVM	0.0398 $\pm$ 0.0081(19)	0.0387 $\pm$ 0.0032(19)	0.3063 $\pm$ 0.0138(20)
PCA	0.0466 $\pm$ 0.0082(14)	0.0432 $\pm$ 0.0026(14)	0.2967 $\pm$ 0.0169(22)
QMCD	0.0594 $\pm$ 0.0072(11)	0.0377 $\pm$ 0.0032(23)	0.3283 $\pm$ 0.0192(9)
SO_GAAL	0.0237 $\pm$ 0.0083(22)	0.0902 $\pm$ 0.0115(4)	0.3660 $\pm$ 0.0508(2)
Sampling	0.1238 $\pm$ 0.0543(3)	0.0420 $\pm$ 0.0030(17)	0.3179 $\pm$ 0.0256(14)
TCCM	0.0562 $\pm$ 0.0186(12)	0.0385 $\pm$ 0.0049(21)	0.3585 $\pm$ 0.0255(4)
VAE	0.0616 $\pm$ 0.0121(10)	0.0462 $\pm$ 0.0036(11)	0.3138 $\pm$ 0.0179(15)

TABLE XXIV: AUPRC, large datasets (1/2).

Model	1_ALOI	8_celeba	10_cover	11_donors	13_fraud	16_http
ABOD	0.1185 $\pm$ 0.0111(2)	0.0202 $\pm$ 0.0005(19)	0.0288 $\pm$ 0.0013(17)	0.0492 $\pm$ 0.0003(19)	0.0159 $\pm$ 0.0011(18)	0.0077 $\pm$ 0.0002(18)
AutoEncoder	0.0419 $\pm$ 0.0033(8)	0.0499 $\pm$ 0.0104(12)	0.4044 $\pm$ 0.0611(3)	0.3215 $\pm$ 0.0309(4)	0.5606 $\pm$ 0.0488(1)	0.7764 $\pm$ 0.1551(6)
CBLOF	0.0406 $\pm$ 0.0034(9)	0.0617 $\pm$ 0.0294(10)	0.0772 $\pm$ 0.0259(10)	0.3004 $\pm$ 0.0451(6)	0.4031 $\pm$ 0.0301(8)	0.4121 $\pm$ 0.0062(13)
DIF	0.0438 $\pm$ 0.0026(5)	0.0491 $\pm$ 0.0041(13)	0.3492 $\pm$ 0.0373(4)	0.2382 $\pm$ 0.0192(9)	0.4106 $\pm$ 0.0472(7)	0.3531 $\pm$ 0.0072(15)
DiffInt	0.0807 $\pm$ 0.0033(4)	0.0528 $\pm$ 0.0056(11)	0.8938 $\pm$ 0.0073(1)	0.9994 $\pm$ 0.0005(1)	0.5503 $\pm$ 0.0193(2)	0.8872 $\pm$ 0.1252(2)
ECOD	0.0335 $\pm$ 0.0014(20)	0.1007 $\pm$ 0.0055(4)	0.1419 $\pm$ 0.0065(7)	0.3098 $\pm$ 0.0022(5)	0.2513 $\pm$ 0.0266(13)	0.2311 $\pm$ 0.0028(16)
GMM	0.0403 $\pm$ 0.0025(11)	0.0910 $\pm$ 0.0042(6)	0.1114 $\pm$ 0.0025(8)	0.2779 $\pm$ 0.0016(8)	0.4888 $\pm$ 0.0380(4)	0.8511 $\pm$ 0.0052(4)
HBOS	0.0343 $\pm$ 0.0013(17)	0.1009 $\pm$ 0.0056(3)	0.0352 $\pm$ 0.0019(16)	0.1826 $\pm$ 0.0026(14)	0.2132 $\pm$ 0.0351(14)	0.5106 $\pm$ 0.1107(9)
IForest	0.0337 $\pm$ 0.0016(19)	0.0682 $\pm$ 0.0095(9)	0.0384 $\pm$ 0.0049(14)	0.2907 $\pm$ 0.0495(7)	0.1265 $\pm$ 0.0279(17)	0.3906 $\pm$ 0.0891(14)
KDE	0.0423 $\pm$ 0.0034(6)	0.0406 $\pm$ 0.0021(15)	0.0495 $\pm$ 0.0016(12)	-	0.3945 $\pm$ 0.0308(11)	-
KPCA	0.0386 $\pm$ 0.0017(14)	-	-	-	-	-
LMDD	0.0364 $\pm$ 0.0017(16)	-	-	-	-	-
LOF	0.0901 $\pm$ 0.0074(3)	0.0172 $\pm$ 0.0006(20)	0.0136 $\pm$ 0.0012(19)	0.1053 $\pm$ 0.0021(16)	0.0018 $\pm$ 0.0001(19)	0.0254 $\pm$ 0.0038(17)
LUNAR	0.1491 $\pm$ 0.0110(1)	0.0356 $\pm$ 0.0018(16)	0.8678 $\pm$ 0.0130(2)	0.9991 $\pm$ 0.0006(2)	0.4674 $\pm$ 0.0555(5)	0.5629 $\pm$ 0.3048(7)
MCD	0.0341 $\pm$ 0.0017(18)	0.0958 $\pm$ 0.0171(5)	0.0158 $\pm$ 0.0004(18)	0.1983 $\pm$ 0.1187(12)	0.5063 $\pm$ 0.0249(3)	0.8727 $\pm$ 0.0061(3)
MO_GAAL	0.0290 $\pm$ 0.0016(23)	0.0136 $\pm$ 0.0017(21)	0.0101 $\pm$ 0.0008(20)	-	0.0009 $\pm$ 0.0000(20)	-
OCSVM	0.0421 $\pm$ 0.0034(7)	0.0772 $\pm$ 0.0048(8)	0.1093 $\pm$ 0.0046(9)	0.2029 $\pm$ 0.0021(11)	0.3949 $\pm$ 0.0306(10)	0.4138 $\pm$ 0.0067(12)
PCA	0.0380 $\pm$ 0.0018(15)	0.1121 $\pm$ 0.0050(2)	0.0732 $\pm$ 0.0015(11)	0.1673 $\pm$ 0.0012(15)	0.1506 $\pm$ 0.0159(16)	0.4758 $\pm$ 0.0056(10)
QMCD	0.0322 $\pm$ 0.0011(21)	0.0222 $\pm$ 0.0006(17)	0.0378 $\pm$ 0.0013(15)	0.0937 $\pm$ 0.0010(17)	0.2982 $\pm$ 0.0382(12)	0.5394 $\pm$ 0.0175(8)
SO_GAAL	0.0292 $\pm$ 0.0015(22)	0.0214 $\pm$ 0.0010(18)	0.0100 $\pm$ 0.0003(21)	0.0513 $\pm$ 0.0082(18)	0.0009 $\pm$ 0.0001(20)	0.0038 $\pm$ 0.0001(19)
Sampling	0.0400 $\pm$ 0.0036(12)	0.0432 $\pm$ 0.0264(14)	0.0465 $\pm$ 0.0212(13)	0.2270 $\pm$ 0.0496(10)	0.3992 $\pm$ 0.0286(9)	0.4480 $\pm$ 0.1252(11)
TCCM	0.0406 $\pm$ 0.0032(9)	0.0833 $\pm$ 0.0067(7)	0.2212 $\pm$ 0.1091(5)	0.9181 $\pm$ 0.0839(3)	0.4241 $\pm$ 0.0526(6)	0.9790 $\pm$ 0.0097(1)
VAE	0.0393 $\pm$ 0.0021(13)	0.1142 $\pm$ 0.0058(1)	0.1664 $\pm$ 0.0069(6)	0.1888 $\pm$ 0.0413(13)	0.1761 $\pm$ 0.0192(15)	0.8511 $\pm$ 0.0060(4)

TABLE XXV: AUPRC, large datasets (2/2).

Model	22_magic	23_mammography	32_shuttle	33_skin	34_smtp
ABOD	0.7128 $\pm$ 0.0079(9)	0.0240 $\pm$ 0.0016(21)	0.1325 $\pm$ 0.0051(19)	0.1771 $\pm$ 0.0013(21)	0.0014 $\pm$ 0.0002(20)
AutoEncoder	0.7687 $\pm$ 0.0135(3)	0.3620 $\pm$ 0.0625(2)	0.9496 $\pm$ 0.0117(12)	0.4034 $\pm$ 0.1190(10)	0.3247 $\pm$ 0.2130(10)
CBLOF	0.7596 $\pm$ 0.0142(4)	0.0785 $\pm$ 0.0120(19)	0.9627 $\pm$ 0.0060(8)	0.5527 $\pm$ 0.0108(4)	0.4802 $\pm$ 0.1330(6)
DIF	0.7236 $\pm$ 0.0113(7)	0.2561 $\pm$ 0.0457(6)	0.9701 $\pm$ 0.0083(5)	0.4606 $\pm$ 0.0187(8)	0.6147 $\pm$ 0.1393(3)
DiffInt	0.8143 $\pm$ 0.0034(1)	0.3010 $\pm$ 0.0216(4)	0.9911 $\pm$ 0.0027(2)	0.9647 $\pm$ 0.0053(1)	0.7004 $\pm$ 0.0484(1)
ECOD	0.5576 $\pm$ 0.0058(18)	0.4516 $\pm$ 0.0447(1)	0.9559 $\pm$ 0.0078(11)	0.1988 $\pm$ 0.0010(18)	0.5990 $\pm$ 0.1003(4)
GMM	0.7397 $\pm$ 0.0068(5)	0.2793 $\pm$ 0.0472(5)	0.9404 $\pm$ 0.0117(13)	0.4953 $\pm$ 0.0018(5)	0.3155 $\pm$ 0.1184(11)
HBOS	0.6460 $\pm$ 0.0072(14)	0.2150 $\pm$ 0.0363(10)	0.9668 $\pm$ 0.0108(6)	0.3693 $\pm$ 0.0026(11)	0.1901 $\pm$ 0.1312(12)
IForest	0.6996 $\pm$ 0.0167(10)	0.2438 $\pm$ 0.0504(9)	0.9765 $\pm$ 0.0090(4)	0.4900 $\pm$ 0.0084(6)	0.0054 $\pm$ 0.0008(19)
KDE	0.6595 $\pm$ 0.0057(11)	0.1058 $\pm$ 0.0136(15)	0.9596 $\pm$ 0.0069(10)	0.2953 $\pm$ 0.0015(13)	0.1038 $\pm$ 0.0424(15)
KPCA	0.4320 $\pm$ 0.0090(20)	0.0744 $\pm$ 0.0108(20)	0.4836 $\pm$ 0.0176(18)	-	-
LMDD	0.3822 $\pm$ 0.0185(21)	0.0937 $\pm$ 0.0280(16)	0.1166 $\pm$ 0.0400(20)	0.1890 $\pm$ 0.0482(20)	0.0717 $\pm$ 0.0759(16)
LOF	0.5361 $\pm$ 0.0044(19)	0.1153 $\pm$ 0.0104(13)	0.0921 $\pm$ 0.0057(21)	0.2298 $\pm$ 0.0015(17)	0.1215 $\pm$ 0.0900(14)
LUNAR	0.8041 $\pm$ 0.0064(2)	0.3489 $\pm$ 0.0518(3)	0.9939 $\pm$ 0.0074(1)	0.9477 $\pm$ 0.0082(2)	0.6708 $\pm$ 0.0870(2)
MCD	0.6573 $\pm$ 0.0060(12)	0.1425 $\pm$ 0.1684(12)	0.8408 $\pm$ 0.0123(16)	0.4762 $\pm$ 0.0021(7)	0.0059 $\pm$ 0.0012(18)
MO_GAAL	0.2282 $\pm$ 0.0055(23)	0.0128 $\pm$ 0.0009(22)	0.0371 $\pm$ 0.0007(22)	0.1939 $\pm$ 0.0307(19)	0.0003 $\pm$ 0.0001(21)
OCSVM	0.6065 $\pm$ 0.0048(16)	0.1124 $\pm$ 0.0148(14)	0.9597 $\pm$ 0.0069(9)	0.3467 $\pm$ 0.0021(12)	0.1629 $\pm$ 0.0699(13)
PCA	0.5939 $\pm$ 0.0065(17)	0.1989 $\pm$ 0.0247(11)	0.9164 $\pm$ 0.0063(15)	0.1632 $\pm$ 0.0009(22)	0.4109 $\pm$ 0.1316(9)
QMCD	0.6329 $\pm$ 0.0082(15)	0.0840 $\pm$ 0.0156(17)	0.6512 $\pm$ 0.0358(17)	0.2299 $\pm$ 0.0011(16)	0.0222 $\pm$ 0.0313(17)
SO_GAAL	0.2286 $\pm$ 0.0060(22)	0.0128 $\pm$ 0.0009(22)	0.0369 $\pm$ 0.0005(23)	0.2632 $\pm$ 0.0935(14)	0.0003 $\pm$ 0.0001(21)
Sampling	0.7256 $\pm$ 0.0190(6)	0.0840 $\pm$ 0.0224(17)	0.9660 $\pm$ 0.0072(7)	0.5764 $\pm$ 0.0818(3)	0.4462 $\pm$ 0.1391(7)
TCCM	0.7154 $\pm$ 0.0256(8)	0.2440 $\pm$ 0.0541(8)	0.9840 $\pm$ 0.0041(3)	0.4232 $\pm$ 0.2331(9)	0.5273 $\pm$ 0.1615(5)
VAE	0.6508 $\pm$ 0.0137(13)	0.2533 $\pm$ 0.0474(7)	0.9336 $\pm$ 0.0114(14)	0.2381 $\pm$ 0.0014(15)	0.4244 $\pm$ 0.1402(8)

TABLE XXVI: AUPRC, high-dimensional datasets (1/2).

Model	3_backdoor	5_campaign	9_census	17_InternetAds	24_mnist	25_musk
ABOD	0.0506 $\pm$ 0.0021(17)	0.1690 $\pm$ 0.0041(17)	0.0467 $\pm$ 0.0004(10)	0.2707 $\pm$ 0.0321(17)	0.2124 $\pm$ 0.0143(15)	0.0349 $\pm$ 0.0173(18)
AutoEncoder	0.8526 $\pm$ 0.0112(4)	0.3426 $\pm$ 0.0116(6)	-	0.8064 $\pm$ 0.0288(2)	0.5906 $\pm$ 0.0384(3)	1.0000 $\pm$ 0.0000(1)
CBLOF	0.0859 $\pm$ 0.0166(14)	0.2479 $\pm$ 0.0200(15)	-	0.5940 $\pm$ 0.0186(7)	0.4009 $\pm$ 0.0413(10)	1.0000 $\pm$ 0.0000(1)
DIF	0.7901 $\pm$ 0.0356(7)	0.2486 $\pm$ 0.0192(14)	-	0.2758 $\pm$ 0.0231(16)	0.4513 $\pm$ 0.0344(9)	0.9963 $\pm$ 0.0040(11)
DiffInt	0.8897 $\pm$ 0.0092(2)	0.3698 $\pm$ 0.0056(2)	0.0767 $\pm$ 0.0005(5)	0.5616 $\pm$ 0.0068(10)	0.5209 $\pm$ 0.0264(6)	0.9998 $\pm$ 0.0025(10)
ECOD	0.1094 $\pm$ 0.0036(13)	0.3696 $\pm$ 0.0115(3)	-	0.5749 $\pm$ 0.0143(9)	0.2070 $\pm$ 0.0118(16)	0.9606 $\pm$ 0.0175(14)
GMM	0.8557 $\pm$ 0.0138(3)	0.3608 $\pm$ 0.0071(4)	-	0.8295 $\pm$ 0.0217(1)	0.5760 $\pm$ 0.0312(4)	1.0000 $\pm$ 0.0000(1)
HBOS	0.0380 $\pm$ 0.0011(19)	0.3864 $\pm$ 0.0133(1)	-	0.2541 $\pm$ 0.0652(18)	0.1543 $\pm$ 0.0124(18)	1.0000 $\pm$ 0.0000(1)
IForest	0.0522 $\pm$ 0.0083(15)	0.3123 $\pm$ 0.0210(10)	0.0746 $\pm$ 0.0029(6)	0.1604 $\pm$ 0.0224(23)	0.3811 $\pm$ 0.0537(11)	0.3308 $\pm$ 0.1853(16)
KDE	0.7243 $\pm$ 0.0222(8)	0.2834 $\pm$ 0.0111(13)	-	0.7507 $\pm$ 0.0180(5)	0.6664 $\pm$ 0.0270(2)	1.0000 $\pm$ 0.0000(1)
KPCA	-	0.0878 $\pm$ 0.0021(23)	-	0.2184 $\pm$ 0.0346(20)	0.0939 $\pm$ 0.0064(21)	0.0376 $\pm$ 0.0149(17)
LMDD	-	0.1092 $\pm$ 0.0043(20)	-	0.4432 $\pm$ 0.0183(13)	0.0856 $\pm$ 0.0057(23)	0.0299 $\pm$ 0.0031(19)
LOF	0.2406 $\pm$ 0.0143(9)	0.1358 $\pm$ 0.0030(18)	0.0613 $\pm$ 0.0005(8)	0.4235 $\pm$ 0.0355(14)	0.2629 $\pm$ 0.0225(14)	0.0259 $\pm$ 0.0051(20)
LUNAR	0.8908 $\pm$ 0.0123(1)	0.2889 $\pm$ 0.0122(12)	-	0.8010 $\pm$ 0.0240(4)	0.6665 $\pm$ 0.0287(1)	1.0000 $\pm$ 0.0000(1)
MCD	0.2331 $\pm$ 0.1673(10)	0.3155 $\pm$ 0.0121(9)	0.1879 $\pm$ 0.0548(1)	0.2296 $\pm$ 0.0375(19)	0.3740 $\pm$ 0.1058(12)	0.9951 $\pm$ 0.0116(12)
MO_GAAL	0.0519 $\pm$ 0.0835(16)	0.0998 $\pm$ 0.0183(22)	-	0.1865 $\pm$ 0.0175(22)	0.1068 $\pm$ 0.0420(20)	0.0199 $\pm$ 0.0031(22)
OCSVM	0.1114 $\pm$ 0.0038(12)	0.3116 $\pm$ 0.0120(11)	-	0.5813 $\pm$ 0.0144(8)	0.4554 $\pm$ 0.0251(8)	0.8724 $\pm$ 0.0631(15)
PCA	0.0246 $\pm$ 0.0007(20)	0.1134 $\pm$ 0.0017(19)	0.0617 $\pm$ 0.0005(7)	0.2987 $\pm$ 0.0112(15)	0.0922 $\pm$ 0.0043(22)	0.9999 $\pm$ 0.0004(9)
QMCD	0.0128 $\pm$ 0.0004(21)	0.3406 $\pm$ 0.0152(7)	0.1194 $\pm$ 0.0013(2)	0.7418 $\pm$ 0.0328(6)	0.1640 $\pm$ 0.0134(17)	0.0159 $\pm$ 0.0016(23)
SO_GAAL	0.0443 $\pm$ 0.0635(18)	0.1026 $\pm$ 0.0209(21)	0.0600 $\pm$ 0.0038(9)	0.1898 $\pm$ 0.0100(21)	0.1099 $\pm$ 0.0404(19)	0.0203 $\pm$ 0.0033(21)
Sampling	0.1406 $\pm$ 0.1835(11)	0.2230 $\pm$ 0.0340(16)	-	0.5368 $\pm$ 0.0269(12)	0.3388 $\pm$ 0.1105(13)	0.9913 $\pm$ 0.0105(13)
TCCM	0.8419 $\pm$ 0.0155(6)	0.3291 $\pm$ 0.0169(8)	0.0950 $\pm$ 0.0033(4)	0.5548 $\pm$ 0.0152(11)	0.5063 $\pm$ 0.0472(7)	1.0000 $\pm$ 0.0000(1)
VAE	0.8466 $\pm$ 0.0085(5)	0.3510 $\pm$ 0.0091(5)	0.1119 $\pm$ 0.0020(3)	0.8043 $\pm$ 0.0291(3)	0.5727 $\pm$ 0.0339(5)	1.0000 $\pm$ 0.0000(1)

TABLE XXVII: AUPRC, high-dimensional datasets (2/2).

Model	26_optdigits	35_SpamBase	36_speech	arrhythmia
ABOD	0.0562 $\pm$ 0.0392(12)	0.3895 $\pm$ 0.0078(20)	0.0247 $\pm$ 0.0103(8)	0.3791 $\pm$ 0.0363(18)
AutoEncoder	0.1018 $\pm$ 0.0108(7)	0.7571 $\pm$ 0.0189(2)	0.0220 $\pm$ 0.0086(13)	0.4081 $\pm$ 0.0651(15)
CBLOF	0.1253 $\pm$ 0.0305(6)	0.4588 $\pm$ 0.0198(12)	0.0217 $\pm$ 0.0084(15)	0.4896 $\pm$ 0.0815(8)
DIF	0.0567 $\pm$ 0.0228(11)	0.4532 $\pm$ 0.0279(14)	0.0161 $\pm$ 0.0048(20)	0.4995 $\pm$ 0.0716(7)
DiffInt	0.3739 $\pm$ 0.0616(3)	0.5853 $\pm$ 0.0083(11)	0.0233 $\pm$ 0.0046(10)	0.5067 $\pm$ 0.0406(5)
ECOD	0.0385 $\pm$ 0.0051(17)	0.6205 $\pm$ 0.0105(10)	0.0227 $\pm$ 0.0094(11)	0.5172 $\pm$ 0.0701(3)
GMM	0.0719 $\pm$ 0.0093(10)	0.7368 $\pm$ 0.0186(5)	0.0329 $\pm$ 0.0182(2)	0.3955 $\pm$ 0.0553(17)
HBOS	0.1829 $\pm$ 0.0261(5)	0.7049 $\pm$ 0.0365(8)	0.0250 $\pm$ 0.0114(5)	0.5092 $\pm$ 0.0746(4)
IForest	0.0992 $\pm$ 0.0195(9)	0.7811 $\pm$ 0.0239(1)	0.0244 $\pm$ 0.0150(9)	0.5250 $\pm$ 0.0838(2)
KDE	0.8656 $\pm$ 0.0478(1)	0.4445 $\pm$ 0.0127(17)	0.0465 $\pm$ 0.0207(1)	0.5008 $\pm$ 0.0690(6)
KPCA	0.0317 $\pm$ 0.0060(18)	0.4376 $\pm$ 0.0188(18)	0.0197 $\pm$ 0.0073(19)	0.1603 $\pm$ 0.0393(21)
LMDD	0.0273 $\pm$ 0.0036(20)	0.4499 $\pm$ 0.0337(16)	0.0199 $\pm$ 0.0056(18)	0.2404 $\pm$ 0.0510(19)
LOF	0.0524 $\pm$ 0.0200(13)	0.3528 $\pm$ 0.0058(21)	0.0249 $\pm$ 0.0111(6)	0.4532 $\pm$ 0.0668(12)
LUNAR	0.8328 $\pm$ 0.0875(2)	0.7526 $\pm$ 0.0346(3)	0.0311 $\pm$ 0.0148(3)	0.5266 $\pm$ 0.0686(1)
MCD	0.0388 $\pm$ 0.0063(16)	0.7187 $\pm$ 0.0498(7)	0.0248 $\pm$ 0.0108(7)	0.4173 $\pm$ 0.0752(14)
MO_GAAL	0.0237 $\pm$ 0.0072(22)	0.3057 $\pm$ 0.0269(22)	0.0143 $\pm$ 0.0030(22)	0.1156 $\pm$ 0.0318(22)
OCSVM	0.0401 $\pm$ 0.0056(15)	0.4521 $\pm$ 0.0128(15)	0.0218 $\pm$ 0.0088(14)	0.4846 $\pm$ 0.0678(10)
PCA	0.0298 $\pm$ 0.0043(19)	0.4149 $\pm$ 0.0104(19)	0.0216 $\pm$ 0.0085(16)	0.4433 $\pm$ 0.0536(13)
QMCD	0.0163 $\pm$ 0.0024(23)	0.6210 $\pm$ 0.0247(9)	0.0133 $\pm$ 0.0024(23)	0.2007 $\pm$ 0.0700(20)
SO_GAAL	0.0240 $\pm$ 0.0080(21)	0.3042 $\pm$ 0.0237(23)	0.0144 $\pm$ 0.0036(21)	0.1119 $\pm$ 0.0279(23)
Sampling	0.0996 $\pm$ 0.0674(8)	0.4545 $\pm$ 0.0162(13)	0.0225 $\pm$ 0.0109(12)	0.4849 $\pm$ 0.0726(9)
TCCM	0.2340 $\pm$ 0.1204(4)	0.7235 $\pm$ 0.0224(6)	0.0284 $\pm$ 0.0127(4)	0.4658 $\pm$ 0.0748(11)
VAE	0.0403 $\pm$ 0.0077(14)	0.7460 $\pm$ 0.0198(4)	0.0216 $\pm$ 0.0085(16)	0.4004 $\pm$ 0.0621(16)